# Chapter 18. Noninvasive Positive-Pressure Ventilation

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## Noninvasive Positive-Pressure Ventilation: Introduction

*Noninvasive ventilation* (NIV) refers to the provision of mechanical ventilation without the need for an invasive artificial airway. Many different approaches to assisting ventilation noninvasively have been used in the past, including negative-pressure ventilators, pneumobelts, and rocking beds (see Chapters 16 and 17).[1] By virtue of its effectiveness and convenience compared with other noninvasive approaches, however, noninvasive positive-pressure ventilation (NIPPV) using a mask (or *interface*) that conducts gas from a positive-pressure ventilator into the nose or mouth has become the predominant means of administering NIV throughout the world. NIPPV has long been used to treat chronic respiratory failure caused by chest wall deformities, slowly progressive neuromuscular disorders, or central hypoventilation.[2] In more recent years, NIPPV has been increasingly used to treat patients with various forms of acute respiratory failure.[3] For the purposes of this discussion, NIPPV refers to active ventilator assistance achieved by the noninvasive provision of a mechanical positive-pressure breath during inhalation, and *continuous positive airway pressure* (CPAP) refers to the provision of a nonfluctuating positive-pressure. This chapter discusses the rationale for use, evidence for efficacy of noninvasive positive-pressure techniques in both acute and chronic settings, selection of appropriate patients, techniques for administration, and pitfalls and complications.

## Why Use Noninvasive Ventilation and How Does It Work?

### Rationale

NIV has become an integral component of ventilator support in both acute and chronic settings because it avoids the complications of invasive ventilation. Invasive mechanical ventilation is highly effective and reliable in supporting alveolar ventilation, but endotracheal intubation carries well-known risks of complications that have been described elsewhere in detail (see Chapter 39).[4,5] These complications have been lumped into three main categories: complications related to insertion of the tube and mechanical ventilation, those caused by loss of airway defense mechanisms, and those that occur after removal of the endotracheal tube.[4]

In the first category, aspiration of gastric contents; trauma to the teeth, hypopharynx, esophagus, larynx, and trachea; arrhythmias; hypotension; and barotrauma may occur during placement of a translaryngeal tube.[6,7] Tracheostomy tube placement incurs risks of hemorrhage, stomal infection, intubation of a false lumen, mediastinitis, and acute injury to the trachea and surrounding structures, including tracheal rupture,[8] and esophageal and vascular injury.[6] In the second category, endotracheal tubes serve as a source of continual irritation, interfere with airway ciliary function, and require frequent suctioning that contributes to airway injury, patient discomfort, and mucus hypersecretion. They also provide a direct channel to the lower airways for microorganisms and other foreign materials, leading to biofilm formation, chronic bacterial colonization, and ongoing inflammation. As a consequence, health care–acquired pneumonias are seen in up to 20% of mechanically ventilated intensive care unit (ICU) patients (see Chapter 46),[9] and sinusitis is seen in 5% to 25% of nasally intubated patients, related to blockade of the sinus ostia and accumulation of infected secretions in the paranasal sinuses (see Chapter 47).[10] In the third category, hoarseness, sore throat, cough, sputum production, hemoptysis, upper airway obstruction secondary to vocal cord dysfunction or laryngeal swelling, and tracheal stenosis all may follow extubation.[11]

In addition, from the point of view of the patient, translaryngeal intubation is uncomfortable and compromises the ability to eat and communicate, contributing to feelings of powerlessness, isolation, and anxiety.[12] This increases the need for sedation, delaying weaning, prolonging the duration of invasive mechanical ventilation, and potentiating the risks of further complications. If tracheostomy placement becomes necessary, sophisticated equipment including suctioning paraphernalia and a high level of technical expertise among caregivers are required. Tracheostomies promote upper airway colonization with gram-negative bacteria, increasing the risk of pneumonias.[7] Furthermore, long-term tracheostomies are complicated by tracheomalacia, granulation tissue formation, and tracheal stenoses that sometimes obstruct the airway, chronic pain, and tracheoesophageal or even tracheoarterial fistulas.[11] These considerations and potential complications may limit the options for chronic care placement, add substantially to the costs of care,[13] and even preclude home discharge in patients with limited caregiver and financial resources.

NIV can avoid many of these complications if candidates are selected properly using guidelines that are discussed in detail later. NIV leaves the upper airway intact, preserves airway defense mechanisms, and allows patients to eat, drink, verbalize, and expectorate secretions. NIPPV reduces the infectious complications of mechanical ventilation, including nosocomial pneumonia and sinusitis.[14,15] It also enhances patient comfort, convenience, and mobility at no greater[16] or even less[13] cost than endotracheal intubation. NIV can be administered outside the ICU setting as long as adequate nursing and respiratory therapy support can be provided, allowing for more rational use of acute care beds, and it greatly simplifies care for patients with chronic respiratory failure in the home.

## Mechanisms of Noninvasive Positive-Pressure Ventilation Action

### Acute Respiratory Failure

NIPPV improves the respiratory status of failing patients via a number of mechanisms. Most important, NIPPV reduces the work of breathing by the same mechanism as invasive positive-pressure ventilation: By applying supra-atmospheric pressure intermittently to the airways, it increases transpulmonary pressure, inflates the lungs, augments tidal volume, and unloads the inspiratory muscles. Exhalation is achieved by passive lung recoil. Studies in patients with severe stable chronic obstructive pulmonary disease (COPD) or restrictive thoracic disorders show that NIPPV reduces or, if inflation pressure is sufficient, even eliminates diaphragmatic work.[17,18] In patients with severe COPD exacerbations, the addition of positive end-expiratory pressure (PEEP) to inspiratory pressure support further reduces the work of breathing by counteracting the effects of auto-PEEP. This combination (pressure support plus PEEP) lowers diaphragmatic pressure swings even more than with either pressure support or PEEP alone.[19] These actions lead to a prompt reduction in respiratory rate, sternocleidomastoid muscle activity, dyspnea, and carbon dioxide ($CO_2$) retention.

Other beneficial actions include an increase in functional residual capacity that opens collapsed alveoli, reducing shunt and enhancing ventilation–perfusion ratios in certain forms of respiratory failure, such as acute cardiogenic pulmonary edema. These effects improve oxygenation and may further reduce the work of breathing because the respiratory system is shifted to a more compliant position on its pressure-volume curve. In addition, CPAP alone (and with NIPPV) may improve left-ventricular function by virtue of an afterload-reducing effect of increased intrathoracic pressure.[20] This effect occurs mainly in patients with dilated, hypocontractile left ventricles whose heart function is more dependent on afterload than on preload. The increased intrathoracic pressure reduces both preload and afterload, but the latter effect predominates, lowering transmyocardial pressure and enhancing cardiac output.[21]

A major effect of NIPPV that appears to be responsible for benefits reported in many studies, including reduced complication rates, mortality, and hospital lengths of stay, is a reduction in health care–acquired infections. Two prospective surveys[14,15] observed roughly a fourfold reduction in the risk of health care–acquired pneumonia compared with physiologically matched endotracheally intubated patients, even after controlling for severity of illness. Patients treated with NIPPV also tend to receive fewer other invasive interventions, such as urinary bladder catheters or central intravenous lines,[15] and this also likely contributes to a lower rate of health care–acquired infections and episodes of sepsis.

Despite the advantages of NIPPV related to the avoidance of airway invasion, the lack of a direct connection to the lower airway also poses a number of challenges. The patient must be able to protect his or her airway and clear secretions adequately, or failure is inevitable. The patient's upper airway must permit airflow into the lungs, so NIPPV cannot be used in patients with high-grade, fixed, upper-airway obstructions. In addition, air leaks around the interface seal or via other routes are nearly ubiquitous with NIPPV and may interfere with the efficacy of ventilation. Furthermore, the patient must be able to cooperate and synchronize breathing with the ventilator, or no reduction in the work of breathing can be achieved, but the patient cannot be heavily sedated or paralyzed to achieve synchrony. Thus, patients who are to receive NIPPV must be selected carefully and managed with an eye to these limitations in ways that differ from the approach used for invasive mechanical ventilation.

### Chronic Respiratory Failure

Long-term NIPPV is used mainly nocturnally during sleep, when intermittent air leaking through the mouth or under the mask seal is universal, but sufficient air usually enters the lungs to assist ventilation.[22] The adaptations that permit air entry into the lungs while NIPPV is administered during sleep are poorly understood, but resistance to airflow in the upper airway is undoubtedly an important factor. In one study, large amounts of air leaking through the mouth during nasal CPAP increased nasal resistance,[23] an effect that was countered by provision of heated, humidified air, consistent with the idea that nasal mucosal cooling was responsible. Increases in nasal resistance secondary to this mechanism, upper airway infection, or allergy is likely to reduce delivered tidal volumes during nasal NIPPV. Passive positioning of the soft palate is also important in maintaining patency of the upper airway,[24] as underlined by the observation that patients treated with nasal CPAP experience increased air leaking through the mouth after uvulopharyngoplasty.[25]

Glottic aperture is also important in determining the flow of gas into the lower airways during NIPPV. Compared with the awake state, the glottic aperture narrows and delivered tidal volume falls when NIPPV is administered during stage 1 or 2 sleep.[26,27] In deeper sleep (stage 3 or 4), the glottic aperture widens, permitting more ventilation; if minute volume is increased excessively, however, the glottic aperture narrows once again, partly related to the reduction in the partial pressure of arterial carbon dioxide ($Pa_{CO_2}$). These findings indicate that both sleep stage and the amount of ventilator assistance influence glottic aperture, which is a potentially important determinant of the efficacy of NIV. They also apply mainly to controlled modes of ventilation;[28] glottic aperture is not as important when NIPPV is administered via a pressure-limited "bilevel" ventilator in the spontaneous mode.[29]

Three theories have been proposed to explain the mechanism by which stabilization of daytime gas exchange is achieved in patients with chronic respiratory failure who are receiving ventilator assistance for as little as 4 to 6 hours nightly. One postulates that NIV rests chronically fatigued respiratory muscles, thereby improving daytime respiratory muscle function.[30,31] Supporting this theory are studies demonstrating that respiratory muscles do indeed rest during NIV;[32–34] also, indices of respiratory muscle strength and endurance may improve in patients with chronic respiratory failure after varying periods of noninvasive ventilatory assistance.[32–35] Conversely, chronic respiratory muscle fatigue has never been defined adequately or demonstrated convincingly; other studies have failed to demonstrate improvement in respiratory muscle function after initiation of NIV,[36] and some studies have demonstrated that patients with neuromuscular disease have stable $Pa_{CO_2}$ values for years despite a progressive decline in pulmonary function.[37]

A second theory proposes that NIV improves respiratory system compliance by reversing microatelectasis of the lung, thereby diminishing daytime work of breathing.[38] This theory derives from studies showing improvements in forced vital capacity (FVC) without changes in indices of respiratory muscle strength after periods of positive-pressure ventilation. Once again, however, data are conflicting, with a number of studies showing no changes in vital capacity after periods of NIV.[35,36] In addition, computed tomographic scanning of the chest indicates that microatelectasis is not an important contributor to chest wall restriction in patients with respiratory muscle weakness.

A third theory proposes that NIV lowers the respiratory center "set point" for $CO_2$ by reversing chronic hypoventilation.[30,39] During deeper stages of sleep, particularly rapid eye movement sleep, upper airway muscle tone and the activity of nondiaphragmatic inspiratory muscles diminish.[40] This response may be exaggerated in patients with ventilatory impairment, leading to progressive nocturnal hypoventilation. Repeated episodes of nocturnal hypoventilation are thought to lead to a gradual accumulation of bicarbonate, desensitization of the respiratory center to $CO_2$, and worsening of daytime hypoventilation.[39] Nocturnal ventilator assistance reverses nocturnal hypoventilation and allows excretion of bicarbonate and a gradual downward resetting of the respiratory center set point for $CO_2$, thereby reducing daytime hypercarbia. In addition, NIPPV may improve the quantity and quality of sleep by preventing hypoventilation-related arousals that lead to sleep fragmentation,[41] reducing fatigue, and improving daytime function. Evidence for this theory derives from studies showing that when ventilator assistance is discontinued for a night in patients with chronic respiratory failure who have been using nightly NIV, the degree of nocturnal hypoventilation is less than before initiation, suggesting a resetting of respiratory center sensitivity for $CO_2$.[42] Also, nocturnal ventilation, oxygen ($O_2$) saturation, sleep quality, and daytime symptoms deteriorate without reductions in respiratory muscle strength or vital capacity when nocturnal NIPPV is discontinued temporarily in patients with restrictive thoracic disease and improve promptly when NIPPV is resumed.[35,41] Moreover, in sixteen patients with chronic respiratory failure secondary to restrictive thoracic disorders followed prospectively for 3 years after starting NIPPV, $Pa_{CO_2}$ improved in association with an increase in the slope of the ventilatory response curve, whereas the maximal inspiratory pressure remained unchanged.[43] These studies suggest that amelioration of nocturnal hypoventilation with resetting of respiratory center $CO_2$ sensitivity and improved sleep quality may be the most important mechanisms contributing to the efficacy of long-term NIPPV. The three theories, however, are not mutually exclusive, and all could contribute more or less, depending on the patient.

Clearly, much remains to be learned regarding specific mechanisms of action of NIV. Understanding of these mechanisms is complicated by the application of NIV in both acute and chronic settings using many different techniques for patients with varying etiologies of respiratory failure. The ability to unload respiratory muscles appears to be key, particularly in the acute setting. Mechanisms controlling upper airway responses and respiratory center adaptations are less well understood but appear to be critical to success in the long-term setting.

## Epidemiology of Noninvasive Positive-Pressure Ventilation

Acute care applications of NIPPV are increasing in Europe and North America.[44,45] An observational study of NIV utilization for COPD and cardiogenic pulmonary edema patients in acute respiratory failure in a single twenty-six-bed French ICU revealed an increase from 20% of ventilator initiations in 1994 to nearly 90% in 2001.[45] In association with this increase, the occurrence of health care–acquired pneumonias and ICU mortality fell from 20% and 21% to 8% and 7%, respectively. In an Italian study examining outcomes of NIPPV in two different time periods during the 1990s, success rates remained steady despite an increase in acuity of illness scores, suggesting that sicker patients in the later time period were being managed as successfully as less ill patients in the earlier period. Both groups of authors speculated that increased experience and skill of the caregivers was responsible for the increased use and improved outcomes.[44,45]

Sequential surveys of European (mainly French) ICUs demonstrated an increase in the use of NIV as a percentage of total initiations of ventilations from 16% in 1997 to 23% in 2002, with utilization in patients with COPD and cardiogenic pulmonary edema increasing from 50% to 66% and from 38% to 47%, respectively.[46] Esteban et al conducted a worldwide survey in more than twenty countries that compared the trends of mechanical ventilation use and demographics between 1998 and 2004, enrolling more than 1600 patients and showing an increase in NIPPV use from 4.4% to 11.1%[47] The differences in rates between the French ICU survey and the worldwide survey may reflect lower utilization rates in some countries as well as the differing methodologies between the two surveys. The French survey examined use in terms of incidents whereas the worldwide survey focused on prevalence of use. Considering that NIPPV is used for shorter periods of time on average than invasive ventilation, prevalence will accordingly be lower.

Some hospital units lend themselves well to NIPPV applications and have very high utilizations rates. In Italy, Confalonieri et al[48] reported high utilization rates of NIV in specialized respiratory intensive care units, which are similar to "intermediate" or "step-down" units in the United States, where a large proportion of patients have COPD either as an etiology of acute respiratory failure or as a comorbidity. In that setting, 425 of 586 (72.5%) patients requiring mechanical ventilation were treated initially with NIV (374 using NIPPV and fifty-one using an "iron lung").[48]

In a 2003 national audit of COPD exacerbations in the United Kingdom, however, NIV was unavailable in nineteen of 233 hospitals and 39% of ICUs, 36% of "high-dependency units," and 34% of hospital wards.[49] Similar results were seen in a North American survey of use of NIV in seventy-one hospitals in Massachusetts and Rhode Island.[50] Overall use of NIPPV was estimated to be 20% of total ventilator initiations, but 30% of hospitals had estimated rates of less than 15%. Reasons for low utilization were mostly attributed to lack of physician knowledge of NIPPV, inadequate equipment, and lack of staff training. Most disturbingly, estimated use of NIV for COPD exacerbations and cardiogenic pulmonary edema was only 29% and 39% of ventilator initiations, respectively.[50] A follow-up study in Massachusetts using data collected prospectively from 2005 to 2007 revealed an overall 38.7% NIV utilization rate, with 80% and 69% of COPD and cardiogenic pulmonary edema patients, respectively, receiving NIV as the initial mode.[51] More recently, clear evidence for the dramatic increase in NIPPV use derived from a Nationwide Inpatient Survey constituting more than 7 million hospital admissions from 1998 to 2008. NIPPV use increased from 1% to 4.5% of total admissions while there was a concomitant fall in invasive mechanical ventilation from 6% to 3.5%. Mortality rates improved in most groups, but the authors raised concerns about a small subgroup of patients who were transitioned from initial NIPPV to invasive ventilation and had a 29% mortality, higher than in patients who were treated with invasive ventilation from the start.[52]

A national survey of U.S. Veterans Affairs hospitals showed that despite wide availability of NIV, its perceived use was low.[53] Almost two-thirds of respiratory therapists responding to the survey thought that NIV was used less than half of the time when its use was indicated. The survey also revealed that wide variations in the perception of NIV use was dependant partly on the size of the ICUs, with larger ones reporting more frequent use.[53] Along these lines, a Canadian study reported that between 1998 and 2003, only 66% of patients meeting criteria for NIPPV actually received it.[54]

Suboptimal utilization has been reported in non-Western countries as well. A Korean survey reported that NIV was used in just two of twenty-four university hospitals and comprised only 4% of ventilator initiations. A majority of the physician staff (62%) and 42% of the nurses expressed a desire for additional educational programs on NIV.[55] In an Indian survey of 648 physicians, perceived NIV use was mostly limited to the ICU (68.4%), and COPD was the most common indication for its use.[56] Findings of this survey were similar to those of the Korean, European, and North American surveys in that rates of NIV use varied widely between centers, with a substantial portion reporting low rates. Thus, although overall use of NIV is clearly increasing, these findings underline the need for NIV educational programs at individual hospitals that permit caregivers to develop the requisite expertise in administering NIV.

## Noninvasive Positive-Pressure Ventilation in the Acute Care Setting

### Evidence for Efficacy

Numerous acute applications of NIPPV have been described, but only a few are supported by strong evidence (Table 18-1). The following sections discuss important applications according to the type of respiratory failure.

Table 18-1: Types of Acute Respiratory Failure Treated with Noninvasive Ventilation Graded

| | References |
|---|---|
| **A. Strong Evidence—Recommended** | |
| Exacerbation of COPD | 17, 59 to 64, 66 to 70 |
| Acute cardiogenic pulmonary edema | 104 to 126 |
| Immunocompromised (hematologic malignancy, bone marrow or solid-organ transplantation, AIDS) | 140 to 144 |
| Facilitation of weaning/extubation patients with COPD | 168 to 180 |
| **B. Intermediate Evidence—Guideline** | |
| Asthma | 66 to 83 |
| Community-acquired pneumonia in patients with COPD | 134 |
| Extubation failure in patients with COPD | 175, 177 to 180 |
| Hypoxemic respiratory failure | 96 to 100 |
| Do-not-intubate patients (COPD and CHF) | 150 to 157 |
| Postoperative respiratory failure (lung resection, bariatric, CABG) | 158 to 167 |
| **C. Weaker Evidence—Optional** | |
| Acute respiratory distress syndrome (ARDS) with single-organ involvement | 144 to 147 |
| Community-acquired pneumonia (non-COPD) | 135 to 140 |
| Cystic fibrosis | 84 to 86 |
| Facilitation of weaning/extubation failure (non-COPD) | 174, 176 |
| Neuromuscular disease/chest wall deformity | 79, 83, 84 |
| Obstructive sleep apnea/obesity hypoventilation | 88, 89 |
| Trauma | 148, 149 |
| Upper airway obstruction | |
| **D. Not Recommended** | |
| Acute deterioration in end-stage interstitial pulmonary fibrosis | 94, 95 |
| Severe ARDS with multiple organ dysfunction | |

| | References |
|---|---|
| Postoperative upper airway or esophageal surgery | |
| Upper airway obstruction with high risk for occlusion | |

*Abbreviations: AIDS,* acquired immune deficiency syndrome; *CABG,* coronary artery bypass graft; *CHF,* congestive heart failure; *COPD,* chronic obstructive pulmonary disease.

Level of evidence: **A,** multiple randomized, controlled trials and meta-analyses; **B,** single controlled trial and cohort series or multiple randomized studies with conflicting findings; **C,** anecdotal reports and case series; **D,** not recommended based on contrary evidence or expert opinion.

## Obstructive Diseases

### Chronic Obstructive Pulmonary Disease

Patients with exacerbations of COPD usually are good candidates for NIPPV because they respond to partial ventilator support, hypoxemia is usually mild to moderate, and the condition is most often reversible within a few days. Thus numerous earlier uncontrolled studies have reported that NIPPV avoids intubation in patients with COPD, with success rates ranging from 58% to 93%. Some studies have reported the use of CPAP alone to treat acute exacerbations of COPD,[57,58] based on the rationale that by counterbalancing auto-PEEP, it will reduce the work of breathing.[59] In these studies, relatively low levels of nasal CPAP (5 to 9.3 cm $H_2O$) were associated with improvements in $Pa_{CO_2}$ and arterial oxygen tension ($Pa_{O_2}$), and few patients required intubation. The lack of controls, however, renders these studies inconclusive.

Kramer et al[60] randomized thirty-one patients with various etiologies for respiratory failure, twenty-one of whom had COPD, to receive NIPPV or conventional therapy. Among COPD patients who received NIPPV in their study, respiratory rate and $Pa_{CO_2}$ fell more rapidly during the first hour of therapy than among controls, and intubation rates were reduced to 9% compared with 67% in controls. In a multicenter European trial[17] of eighty-five patients with COPD randomized to receive face-mask pressure-support ventilation (PSV) or conventional therapy, respiratory rate but not $Pa_{CO_2}$ fell significantly in the NIPPV group during the first hour; intubation (26% vs. 74%), complication (16% vs. 48%), and mortality (9% vs. 29%) rates and hospital lengths of stay (35 vs. 23 days) all were significantly lower in the NIPPV than in the control group.

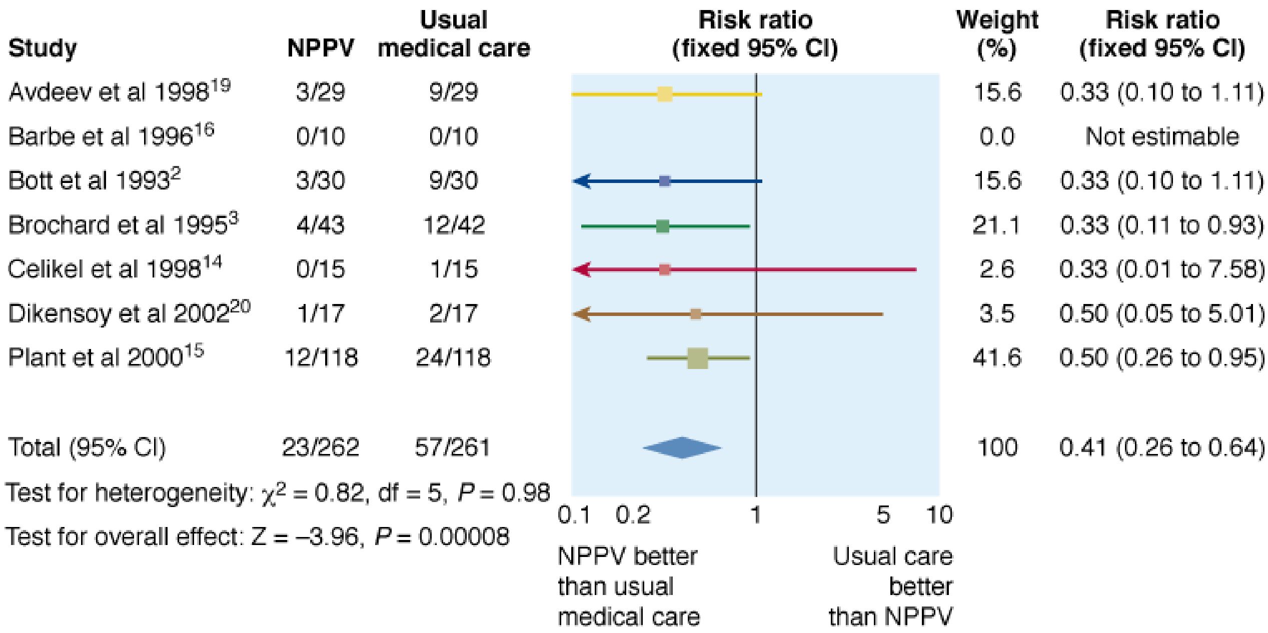
Another randomized, controlled trial compared the efficacy of standard medical therapy with NIPPV in thirty patients with acute hypercapnic respiratory failure caused by exacerbations, pneumonia, or congestive heart failure.[61] Those randomized to NIPPV had greater improvements in pH and respiratory rate within 6 hours, higher success rate (93%), and shorter hospital lengths of stay (11.7 vs. 14.6 days, $p < 0.05$) than controls. The largest study to date on NIPPV for exacerbations of COPD randomized 236 patients to receive NIPPV or standard therapy at fourteen British centers.[62] NIPPV was administered in general respiratory wards by nurses who had a few hours of in-service training with the technique. Patients treated with NIPPV had lower intubation (15% vs. 27%) and mortality (10% vs. 20%) rates than controls, but the benefit was seen only in patients with pH values of 7.3 or greater. The authors concluded that although NIPPV proved to be effective in their study, these sicker patients probably should have been treated in an ICU.

In addition to the favorable findings regarding the use of NIPPV for acute exacerbations of COPD, some studies have found that 1-year survival rates are better and the need for rehospitalization and consumption of ICU beds over the next year are less for patients treated with NIPPV as opposed to conventional therapy.[63,64] Although these latter studies were not randomized, so the results could have reflected a selection bias favoring less ill patients in the NIPPV group, it is also possible that NIPPV avoids late complications of invasive ventilation, such as sustained muscle weakness or swallowing dysfunction.[10]

Among the many controlled and uncontrolled studies examining the efficacy of NIPPV in exacerbations of COPD, only two have obtained unfavorable results. In one,[65] twenty-five of forty-nine consecutive COPD patients with acute exacerbations were treated with nasal NIPPV, and twenty-four were intolerant and served as the "control" group. Blood gases in both groups improved at similar rates, and no differences in outcome were apparent between the two groups. In the second, Barbe et al[66] randomized twenty-four patients with acute exacerbations of COPD to receive nasal NIPPV or standard therapy. Four of fourteen patients randomized to NIPPV were intolerant; among the remaining patients, blood-gas improvements and hospital lengths of stay were similar, and no differences in intubation or mortality rates were apparent, leading the authors to conclude that NIPPV is ineffective in COPD. Both studies, however, enrolled consecutive patients who, on average, had less-severe blood-gas abnormalities than patients enrolled in favorable studies; none of the patients in the study of Barbe et al[66] required intubation, as did almost three-quarters of the controls in the studies of Kramer et al[60] and Brochard et al.[17] These observations support the contention that the patients in the two unfavorable studies were less ill than those in the favorable studies, and argue that NIPPV should be reserved for sicker patients with COPD who are at risk of requiring intubation.

Multiple randomized, controlled trials lend themselves to meta-analysis. An earlier meta-analysis by Keenan et al[67] concluded that NIPPV significantly reduces mortality and reduces the cost of hospitalization by an average of $3244 (Canadian) compared with conventional therapy. Peter et al[68] examined both COPD and non-COPD causes of acute respiratory failure in their meta-analysis, and concluded that NIPPV significantly reduces the need for intubation as well as mortality. Meta-analyses by Keenan et al[69] and Lightowler et al[70] (Cochrane analysis) observed absolute and relative risk reductions of 28% and 0.42 for intubation, 10% and 0.41 for mortality, and 4.57 and 3.21 hospital days, respectively (all $p < 0.05$) (Fig. 18-1). The analysis of Keenan et al also concluded that there is little evidence to support NIPPV use in milder COPD, although they analyzed only two studies of mild patients. The analysis of Lightowler et al also found that $Pa_{CO_2}$, heart rate, and dyspnea scores improved more rapidly than in conventionally treated patients. A more recent metaanalysis by Quon et al[71] derived similar findings, observing reductions of 65% in intubations, 55% in mortality, and 1.9 days in hospital length of stay among COPD patients admitted with exacerbations and treated with NIPPV. These studies lend strong support to the use of NIPPV for patients with COPD in the acute care setting, leading consensus groups to recommend that the modality "be considered" in selected patients,[72] and the Canadian Thoracic Society Guideline committee to recommend that NIV be considered the ventilatory modality of first choice for patients with acute respiratory failure secondary to exacerbations of COPD.[73] The need for careful patient selection cannot be overemphasized (see "Selection Guidelines" below). NIPPV is best used to avoid intubation, not to replace it. Although NIPPV should be viewed as the ventilator therapy of first choice for appropriate COPD patients, those with contraindications to NIPPV should be intubated and ventilated without delay.

**Figure 18-1**






Forest plot of eight randomized, controlled studies on NPPV in patients with acute respiratory failure secondary to COPD. The reduction in mortality rate was consistent among studies. (Used, with permission, from Lightowler. Noninvasive positive pressure ventilation to treat respiratory failure resulting from exacerbations of chronic obstructive pulmonary disease: Cochrane systematic review and meta-analysis. *Br Med J.* 185–189, 2003.)

In view of the idea that NIPPV is best used to avoid intubation, Squadrone et al[74] asked whether it can serve as an alternative to intubation in patients with COPD and advanced acute hypercapnic respiratory failure (pH ≤ 7.25, $Pa_{CO_2} \geq 70$ mm Hg, respiratory rate ≥ 35 breaths/min). Sixty-four such patients had similar mortality rates and hospital lengths of stay but fewer serious complications (mainly infectious) and a trend toward a higher weaning rate at 30 days compared with a historically matched control group of invasively ventilated patients. The authors concluded that NIPPV can be used as an alternative to invasive mechanical ventilation in severely ill patients with COPD, but considering that the failure rate approached two-thirds in the NIPPV group and that the more recent report by Chandra et al[52] raised concerns about this approach, NIPPV should be applied with great caution in such patients. As pointed out previously, the use of historical controls is a serious design limitation that may favor the treatment group.

Few data provide guidance on selecting patients who might benefit from continued use of NIPPV after hospital discharge. In an uncontrolled retrospective study, Tuggey et al[75] found that patients treated with NIV during their acute admissions and sent home with it had many fewer hospital days per year (25 vs. 78 days; $p = 0.004$) and incurred much lower costs per year ($7407 vs. $23,065) after starting domiciliary NIV than before. Despite the small number of patients and uncontrolled design, these results support the idea that domiciliary NIV should be considered in “revolving-door patients” who require repeated hospital admissions and highlight the need for more definitive studies addressing this question.

### Asthma

Although acute asthma would be anticipated to respond favorably to NIPPV because it shares pathophysiologic features with COPD, much less evidence supports this application. One early report described seventeen patients with asthma who had an average initial pH of 7.25, a $Pa_{CO_2}$ of 65 mm Hg, and were treated with face-mask PSV.[76] Only two required intubation (for increasing $Pa_{CO_2}$), average duration of ventilation was 16 hours, and no complications occurred.

More recently, Fernandez et al[77] reported on fifty-eight patients with status asthmaticus, thirty-three of whom were retrospectively deemed candidates for NIPPV because of persisting $CO_2$ retention ($Pa_{CO_2} > 50$ mm Hg) and because they met other clinical criteria.[77] Of these, eleven were intubated according to clinician preference, and twenty-two were managed noninvasively. The noninvasively and invasively treated groups had similar initial $Pa_{CO_2}$ values, which improved less rapidly in the noninvasive group. Only 14% of the NIPPV-treated patients eventually required intubation, and they had shorter ICU and hospital lengths of stay than the intubated group. Thus far, four randomized, controlled trials have been reported. Holley et al[78] were able to enroll only one-tenth of the roughly 350 patients their power analysis had projected. Not surprisingly, their major outcome variable—intubation rate—was not reduced significantly in their NIPPV group (one of nineteen versus two of sixteen in controls), and they observed no deaths. Their major finding was that physicians who a priori believed that NIPPV was effective were less likely to enroll patients in the trial because of concern that the patients might require intubation if randomized to the control group. Soroksky et al[79] randomized thirty-three patients with severe acute asthma (average initial forced expiratory volume in 1 second [$FEV_1$] roughly 33%) to receive NIPPV or sham therapy via a face mask. The NIPPV group had a significantly greater increase in $FEV_1$ within the first hour (53.5% vs. 28.5%) and fewer hospitalizations (three of seventeen versus ten of sixteen) compared with the sham group. Both groups received aerosolized bronchodilators via a nebulizer, not the ventilator. The authors speculated that the greater improvement in airflow in the NIPPV group might be related to a bronchodilator effect of positive-pressure.

The third randomized trial[80] supports the notion that NIPPV may have an initial bronchodilator effect in acute asthma. In this study, forty-four patients with status asthmaticus (mean $FEV_1$ 33% predicted) received "low" NIPPV (inspiratory and expiratory pressure 6 and 4 cm $H_2O$, respectively), "high" NIPPV (inspiratory and expiratory pressure 8 and 6 cm $H_2O$, respectively), or oxygen supplementation in controls. For the first hour, they received hydrocortisone but no bronchodilators. After the initial hour, the "high" NIPPV group had an improvement in $FEV_1$ over baseline of 20%, compared to no improvement in controls ($P < 0.05$).

In the fourth trial,[81] undoubtedly underpowered, fifty-three patients with severe asthma were randomized to NIPPV (inspiratory pressure 12 cm $H_2O$, expiratory pressure 5 cm $H_2O$) plus standard therapy or standard therapy alone. The NIPPV group spent less time in the ICU and used less bronchodilation, but no other significant differences were observed, including rate of improvement in $FEV_1$ or gas exchange, or rate of intubation.

These studies suggest that NIPPV may be effective at improving airflow, correcting gas-exchange abnormalities, avoiding intubation, and reducing the need for hospitalization in patients with acute severe asthma. Published studies, however, are either uncontrolled or underpowered or the findings have not been replicated. A Cochrane analysis concluded that evidence for use of NIPPV for acute asthma was "very promising" but "controversial" and that more controlled studies are needed.[82] Furthermore, medical therapy alone may be quite effective.[83] Lacking more evidence, no firm conclusions can be drawn regarding the relative effectiveness of NIPPV versus conventional therapy in exacerbations of asthma. The British Thoracic Society Standards of Care Committee opined that NIPPV should not be used routinely for acute asthma.[84] Nonetheless, an empiric trial of NIPPV might be considered in patients not responding promptly to standard medical therapy if selected according to commonly used criteria (see "Selection Guidelines" below). Also, more research into the role of positive-pressure in enhancing acute bronchodilation seems warranted.

### Cystic Fibrosis

NIPPV has been used to treat acutely deteriorating patients with end-stage cystic fibrosis. In one study, six patients with $FEV_1$ ranging from 350 to 800 mL and severe acute-on-chronic $CO_2$ retention (initial $Pa_{CO_2}$ ranging from 63 to 112 mm Hg) were treated with NIPPV for periods of 3 to 36 days; four survived until a heart-lung transplantation could be performed.[84] The same investigators reported more recently on 113 patients with cystic fibrosis treated with NIPPV for acute deteriorations.[85] Ninety of these patients (median $FEV_1$/FVC ratio of 0.5) were listed for lung transplantation, twenty-eight survived to transplantation, ten remained on the list at the time of reporting, and the remainder expired. NIPPV improved hypoxia but not hypercapnia. These case series suggest that NIPPV may serve as a rescue therapy to provide a "bridge to transplantation" for patients with acutely deteriorating cystic fibrosis, but control of airway secretions is still a big challenge and mortality is high if the wait for an organ is prolonged beyond a few months.[86]

#### Upper Airway Obstruction

The inappropriate use of NIPPV in patients with tight, fixed upper airway obstruction should be avoided so as not to delay the institution of definitive therapy. In my experience, however, NIPPV can be used to treat patients with reversible upper airway obstruction, such as that caused by glottic edema following extubation, sometimes in combination with aerosolized medication and/or helium–oxygen gas mixture. Although no controlled trials demonstrate the efficacy of this approach in adults, a controlled trial in ten infants with respiratory failure showed that NIPPV[87] and CPAP were equally efficacious in lowering respiratory rate, but NIPPV contributed to patient–ventilator asynchrony. If used, NIPPV should be administered cautiously and monitored closely because these patients are at risk for precipitous deteriorations.

#### Decompensated Obstructive Sleep Apnea or Obesity Hypoventilation Syndrome

Patients with acute-on-chronic respiratory failure caused by sleep apnea syndrome, often in combination with obesity hypoventilation, have been treated successfully with NIPPV and transitioned to CPAP once stabilized,[88] but no controlled trials have evaluated this application. Sturani et al[89] described the successful use of nasal NIPPV administered with the biphasic intermittent positive airway pressure (BiPAP) device (18 cm $H_2O$ inspiratory and 6 cm $H_2O$ expiratory pressures) in five morbidly obese patients (mean body mass index of 50 kg/$m^2$) with severe sleep apnea. Anecdotally, high inflation pressures, sometimes necessitating use of volume-limited ventilators that have greater pressure-generating capabilities than portable pressure-limited ventilators, may be needed because of high respiratory system impedance.

### Restrictive Diseases

Although NIPPV to treat patients with chronic respiratory failure secondary to restrictive thoracic diseases is well accepted (see “Chronic Respiratory Failure” above), it is used for only a small portion of patients admitted to acute care hospitals with respiratory failure. Accordingly, few studies on the management of acute respiratory failure in these patients have been reported. Small uncontrolled series have reported success using NIPPV to alleviate gas-exchange abnormalities and to avoid intubation in patients with acute respiratory failure secondary to neuromuscular disease[90] and kyphoscoliosis.[91] Despite the lack of evidence, the British Thoracic Society Standards of Care Committee considers that NIPPV “is indicated” in patients with acute or acute-on-chronic respiratory failure secondary to restrictive thoracic diseases.[92]

A regimen for managing acute deteriorations in patients with chronic respiratory failure secondary to neuromuscular disease, reported by Bach et al,[93] requires that patients receive NIV at home 24 hours a day during the exacerbation. When $O_2$ saturation falls below 90% as determined by continuous pulse oximetry, airway secretions are removed aggressively using manually assisted coughing and mechanical aids such as the cough insufflator–exsufflator until $O_2$ saturation returns to the 90% range. In this small series of patients, the need for hospitalization was reduced dramatically after institution of the regimen.[79]

Limited information is available on NIPPV therapy for patients with acutely deteriorating restrictive lung diseases such as idiopathic pulmonary fibrosis. Such patients usually fare poorly with mechanical ventilation.[94] On the other hand, in a recent retrospective cohort of eleven patients with idiopathic pulmonary fibrosis and acute respiratory failure treated with NIPPV,[95] five survived the hospitalization and lived for more than 3 months. Thus, some such patients could be considered for NIPPV if they have a possible reversible superimposed condition and are otherwise good NIPPV candidates.

### Hypoxemic Respiratory Failure

Hypoxemic respiratory failure is defined as a $P_{CO_2}/F_{IO_2}$ ratio of less than 200 and a respiratory rate greater than 35 breaths/min; contributing diagnoses include acute pneumonia, acute pulmonary edema, acute respiratory distress syndrome (ARDS), and trauma.[96] It is an extremely broad category of acute respiratory failure. Hence, perhaps not surprisingly, studies of NIPPV to treat it have yielded conflicting results. Meduri et al[97] were the first to report the successful application of NIPPV in such patients. In a randomized trial,[98] the same authors found no benefit of NIPPV over conventional therapy among all entered patients. Initial hypercapnia predicted a favorable outcome: Patients with an initial $Pa_{CO_2}$ of greater than 45 mm Hg had significantly lower intubation and ICU mortality rates and shorter ICU lengths of stay than normocapnic patients. The authors concluded that hypoxemic respiratory failure without $CO_2$ retention responds poorly to NIPPV.

Conversely, Antonelli et al[96] randomized sixty-four patients with hypoxemic respiratory failure to NIPPV or immediate intubation. Improvements in oxygenation were similar in the two groups, and only 31% of the NIPPV-treated patients required intubation. NIPPV-treated patients had significantly fewer septic complications such as pneumonia or sinusitis (3% vs. 31%), and there were trends toward decreased mortality and ICU length of stay (27% vs. 45% and 9 vs. 15 days, respectively) compared with intubated controls. Another randomized, controlled trial of sixty-one patients with various forms of acute respiratory failure found a significantly reduced intubation rate when patients with acute hypoxemic respiratory failure were treated with NIPPV as opposed to conventional therapy (7.5 vs. 22.6 intubations per 100 ICU days); mortality rates, however, were not significantly different.[99]

More recently, Ferrer et al[100] randomized patients with severe hypoxemia (defined as a $P_{CO_2}$ of less than 60 mm Hg or an arterial oxygen saturation [$SaO_2$] of less than 90% on 50% $F_{IO_2}$ for at least 6 to 8 hours) to receive NIPPV or conventional therapy. Intubation rate was decreased from 52% to 25%, the incidence of septic shock was reduced, and both ICU (39% vs. 18%) and 90-day mortality were lower in the NIPPV group than in controls. In contrast to some previous studies, substantial benefit was observed in patients with severe pneumonia, whereas patients with cardiogenic pulmonary edema had no reduction in intubation rate.

The favorable results of these latter studies might be interpreted to show broad support for the use of NIPPV in patients with hypoxemic respiratory failure. In fact, a recent survey of European pulmonologists and anesthesiologists showed that 48% (more pulmonologists than anesthesiologists) considered acute hypoxemic respiratory failure as a preferred indication for NIPPV.[101] A systematic review, however, noted that although intubations in patients with acute hypoxemic respiratory failure seem to be reduced by NIPPV, the heterogeneity between studies precluded any firm conclusions and recommended against the routine use of NIPPV in these patients.[102] Also, when overall results are favorable in a heterogeneous group of patients, it cannot be assumed that each subgroup benefits equally. It is possible that harm to a particular subgroup could be obscured by benefit in other subgroups. The following subsections examine evidence regarding the use of NIPPV in specific subgroups of patients with acute hypoxemic respiratory failure that may be more appropriate to apply to individual patients.

### Acute Cardiogenic Pulmonary Edema

CPAP, although not a form of mechanical ventilatory assistance per se, was described as a treatment for acute pulmonary edema dating back to the 1930s.[103] Over the past 15 years, a number of studies have demonstrated that CPAP (10 to 12.5 cm $H_2O$) is effective in treating acute pulmonary edema. Rasanen et al[194] randomized forty patients with cardiogenic pulmonary edema to either face-mask CPAP or standard medical therapy, and demonstrated that CPAP more rapidly improves oxygenation and respiratory rate. In a study of fifty-five patients with pulmonary edema, Lin and Chang[105] found that those randomized to face-mask CPAP (adjusted to maintain a $Pa_{O_2}$ of 80 mm Hg or greater) had a lower intubation rate (17.5% vs. 42.5%, $p < 0.05$) than conventionally treated controls. Bersten et al[106] and Lin et al[92] subsequently performed randomized studies on thirty-nine and 100 patients, respectively, demonstrating more rapid improvements in respiratory rates and oxygenation and a reduced need for intubation in patients treated with CPAP. The study of Bersten et al[106] also showed a significant reduction in the length of ICU stay among CPAP-treated patients, and the study of Lin et al[107] showed a trend for a lower hospital mortality rate. The average absolute reduction in intubation rate among these studies was 28% (from 47% in controls to 19% for CPAP). These studies provide strong evidence to support the use of CPAP to treat acute cardiogenic edema.

As discussed earlier, the combination of increased inspiratory pressure and positive expiratory pressure (i.e., pressure support plus PEEP or NIPPV) might be expected to reduce work of breathing more effectively than CPAP alone, bringing about more rapid relief of dyspnea and improvement in gas exchange. Thus, more recent studies have focused on the use of NIPPV to treat acute cardiogenic pulmonary edema. One prospective, uncontrolled study found that face-mask PSV improved pulse oximetry, pH, and $Pa_{CO_2}$ within 30 minutes in twenty-nine patients with acute pulmonary edema, only one of whom required intubation.[108] A second prospective, uncontrolled study[109] reported similar effects on gas exchange, but five of twenty-six patients required intubation, and successfully treated patients had higher $Pa_{CO_2}$ (54 vs. 32 mm Hg) and lower creatine phosphokinase levels (176 vs. 1282 IU) (both $p < 0.05$) than failure. Furthermore, four patients died in the first study and five in the second, three and four, respectively, with myocardial infarctions. The authors concluded that NIPPV is a "highly effective technique." The accompanying editorialist, however, cautioned about applying NIPPV to patients with acute myocardial infarctions.[110]

Several randomized, controlled trials have been performed subsequently comparing NIPPV with conventional $O_2$ therapy to treat patients with acute cardiogenic pulmonary edema. Masip et al[111] found that inspiratory and expiratory pressures of 15 and 5 cm $H_2O$, respectively, lowered the intubation rate from 33% in eighteen controls to 5% among twenty-two patients randomized to NIPPV ($p = 0.037$). NIPPV also improved oxygenation more rapidly, but hospital lengths of stay and mortality rates were similar in the two groups. Sharon et al[112] randomized forty patients to receive NIPPV plus low- or high-dose nitroglycerin. The NIPPV group had higher rates of intubation (80% vs. 20%), myocardial infarction (55% vs. 10%), and death (10% vs. none) compared with the nitroglycerin controls (all $p < 0.05$), leading the authors to conclude that NIPPV was less effective and potentially harmful compared with high-dose nitroglycerin. This inference, however, is suspect because the treatments were not comparable, and the inordinately high intubation rate in the NIPPV group (80%) is difficult to explain. In a larger study, Nava et al[113] randomized 130 patients with acute pulmonary edema to receive NIPPV (average: 14.5 cm $H_2O$ of pressure support and 6.1 cm $H_2O$ of PEEP) or $O_2$ therapy; hypercapnic and normocapnic patients were prospectively distributed equally between groups. As in the earlier studies, NIPPV improved oxygenation, respiratory rate, and dyspnea more rapidly than conventional therapy, but mortality and hospital lengths of stay did not differ between the groups. Overall, the rates of intubation were not significantly different (25% in controls vs. 20% for NIPPV); in the hypercapnic subgroup, however, the intubation rate was lower in the NIPPV group than in controls (6% vs. 29%, $p = 0.015$). These studies suggest that NIPPV is effective therapy for acute pulmonary edema, but whether this is true only for hypercapnic patients awaits further evaluation.

The question of whether NIPPV (the combination of pressure support and PEEP) is superior to CPAP alone is important because CPAP can be delivered more simply and less expensively. An earlier randomized trial comparing the two to treat acute pulmonary edema showed significantly more rapid reductions in respiratory rate, dyspnea scores, and hypercapnia in the NIPPV group compared with the CPAP-treated group.[114] The study was stopped prematurely after enrollment of twenty-seven patients, however, because of a greater myocardial infarction rate in the NIPPV group (71% vs. 31% in controls). This difference may have been attributable to unequal randomization because more patients in the NIPPV group presented with chest pain. The results nonetheless raise concerns about the safety of ventilator techniques used to treat acute pulmonary edema.

More recent randomized, controlled trials comparing NIPPV with CPAP have not detected differences in the myocardial infarction rate. Crane et al[115] randomized sixty patients with cardiogenic pulmonary edema to three different therapies: conventional, CPAP (10 cm $H_2O$), or bilevel ventilation (15 cm $H_2O$ inspiratory and 5 cm $H_2O$ expiratory pressures). Treatment success was 15% in the control group, 35% in the CPAP group, and 45% in the bilevel group ($p = 116$). Although myocardial infarction rate did not differ among the groups, hospital mortality was 30% in the control group, 0% in the CPAP group, and 25% in the bilevel group ($p = 0.029$). The difference in mortality was not statistically significant until after the first week of hospitalization, after patients had stopped using the devices.

Several additional studies have randomized patients with acute cardiogenic pulmonary edema to CPAP or noninvasive PSV plus PEEP and have found no differences in myocardial infarction rates.[116,119] Physiologic variables improved equally in both groups, intubation and mortality rates were similar, and troponin I levels and the speed of clinical resolution were nearly identical. In addition to finding no increase in the myocardial infarction rate in NIV-treated patients, these latter studies also found no clear advantage of NIPPV over CPAP alone. These findings must be interpreted with caution, however, because patients with myocardial infarction or acute ischemia were excluded.

In the largest study reported to date, Gray et al[120] randomized 1069 patients with cardiogenic edema to receive CPAP (5 cm $H_2O$) alone, NIPPV (inspiratory and expiratory pressures 8 and 4 cm $H_2O$, respectively) or oxygen plus routine therapy in controls. No differences were noted between the CPAP and NIPPV groups, so they were combined in the analysis. Although dyspnea and pH improved more rapidly in the positive-pressure groups than in controls, no differences were apparent in intubation or mortality rates. The authors concluded that noninvasive positive-pressure treatment of cardiogenic pulmonary edema was useful to alleviate symptoms, but not to improve other outcomes. The very low intubation rate in this study (roughly 3% in all groups), however, was much lower than in most previous studies despite the use of low positive-pressures, suggesting that the patients had relatively mild respiratory compromise. Thus, the study may not be comparable to the earlier studies showing significant avoidance of intubations or even mortality.

A number of meta-analyses[121–126] have concluded that CPAP alone is effective in reducing dyspnea, improving vital signs and gas exchange, and reducing intubation and mortality rates, without any significant effect on myocardial infarction rates. Similar benefits have been attributed to NIPPV, except that improved mortality has been found in only a few,[124–126] probably because there have been fewer studies of NIPPV than on CPAP. These meta-analyses have also compared NIPPV and CPAP, finding no differences between the two modalities with respect to intubation and mortality rates as well as occurrence of myocardial infarction. These findings have held up, even when meta-analyses were performed after publication of the Gray study.[125,126]

Deciding which patients with acute cardiogenic pulmonary edema should receive NIPPV can be challenging because they may respond rapidly to conventional therapy. Using their single-center registry, Masip et al[127] obtained data on eighty conventionally treated patients with cardiogenic pulmonary edema to identify those at risk for intubation. Patients with a pH of less than 7.25 or hypercapnia and a systolic blood pressure of less than 180 mm Hg were found to be at high risk. The authors recommended that such patients should be “promptly considered” for NIV. Because studies have not shown definitively that NIV is more effective than CPAP, however, the most sensible current recommendation is to use CPAP (10 cm $H_2O$) initially, and consider switching to NIPPV if the patient has unrelenting dyspnea or persisting tachypnea or hypercapnia. Furthermore, either CPAP or NIPPV should be used with great caution, if at all, in patients with acute myocardial infarction or active ischemia. These recommendations are in line with those of a Cochrane analysis that deemed NIV, especially CPAP, as “safe and effective” for the treatment of cardiogenic pulmonary edema in adults.[128]

#### Prehospital Use of CPAP for Cardiogenic Pulmonary Edema

An important trend in the application of CPAP for acute cardiogenic pulmonary edema has been the use in the field by ambulance crews to treat patients before hospitalization. The experience with this practice has been favorable thus far. Plaisance et al[129] observed a strong trend for reduced intubation and mortality rates among 124 patients with cardiogenic pulmonary edema randomized to “early” (started immediately on site) versus “late” (delayed by 15 minutes) CPAP (7.5 cm $H_2O$). In another randomized, controlled trial, Thompson et al[130] observed an absolute reduction of 30% in intubation rate (seventeen of thirty-four patients [50%] vs. seven of thirty-five [20%]) and absolute mortality fell by 21% among patients with cardiogenic pulmonary edema treated with CPAP compared to usual therapy with oxygen, including intubation and bag-valve-mask-ventilation if needed.

A pilot study by Duchateau et al[131] reported an improved respiratory status in twelve “do not intubate” patients when offered NIPPV out-of-hospital by emergency medical services. Respiratory rate decreased from 34 to 27 breaths/min ($p = 0.009$) and pulse oximetry improved from 86% to 94% ($p < 0.01$) with only one intolerant patient. These studies suggest that outcomes of patients with cardiogenic pulmonary edema can be improved by very early initiation of noninvasive positive-pressure therapy in the field and adoption of this as a routine practice for emergency medical services seems likely. A recent Cochrane analysis concluded that prehospital NIV “appears to be a safe and feasible therapy” that may lower the need for intubation compared to initiation in the emergency department.[132] The authors of the Cochrane analysis cautioned, however, that the evidence is preliminary and that cost-effectiveness analyses have not been performed.

### Pneumonia

An earlier retrospective study found that acute severe pneumonia is a predictor of NIV failure, perhaps because NIPPV does little to facilitate the clearance of secretions.[133] Confalonieri et al[134] randomized fifty-six patients with severe community-acquired pneumonia to receive NIPPV or standard $O_2$ therapy. The NIPPV group had fewer intubations (21% vs. 50%) and shorter ICU lengths of stay (1.8 vs. 6 days) than controls (both $p < 0.05$). In addition, NIPPV-treated patients with COPD had significantly better survival at 2 months, thought to be related to fewer late complications of intubation. The most important observation, though, was that all the benefit was attributable to the subgroup with COPD, and no clear benefit of NIPPV was seen in patients without COPD patients. More recently, a prospective study on NIPPV to treat patients with severe community-acquired pneumonia but without COPD found that oxygenation and respiratory rates improved initially in twenty-two of twenty-four patients after starting NIPPV, but 66% eventually required intubation.[135] Based on the preceding evidence, initiation of NIPPV is justifiable in appropriate patients with pneumonia and COPD. The benefit of NIPPV in patients with pneumonia but without COPD has not been established. As such, NIPPV should be used selectively and with caution in such patients.

The severe acute respiratory syndrome epidemic was characterized by a high rate of respiratory failure among afflicted individuals, many of whom were otherwise healthy health care workers. Initially, use of NIPPV was discouraged because of concerns about aerosolization and transmission of the highly contagious coronavirus to other health care workers. Two retrospective studies, however, one from Beijing on twenty-eight patients treated with NIPPV[139] and the other from Hong Kong on twenty patients,[140] suggest that NIPPV is effective in avoiding intubation in some patients. Intubation was required in only 33% and 30% of NIPPV-treated patients in the two studies, respectively. Stringent infection-control measures, including the use of a face mask, an inline viral/bacterial filter in the bilevel ventilator tubing, and a high-efficiency particulate accumulator mask by all health care workers having contact with the patients, prevented transmission of severe acute respiratory syndrome to any caregivers. Both studies reported high rates of barotraumas, 22% and 20%, respectively; it was unclear that this was related to NIPPV. Given the lack of controls, these studies cannot be used to assess the efficacy of NIPPV in severe acute respiratory syndrome, although the lack of transmission to health care workers should allay fears about NIPPV spreading the virus so long as stringent isolation and prevention measures are employed.

More recently, the use of NIPPV to treat influenza pneumonia during the H1N1 epidemic was controversial.[138] Based on the 1 m dispersion of aerosol beyond the mask demonstrated during NIPPV applied to a human-like mannequin,[139] some advised against use of NIPPV for influenza pneumonia, although actual transmission by this route was never demonstrated, nor was it compared to dispersion occurring during invasive mechanical ventilation or even spontaneous breathing.

### Immunocompromised Patients

The use of NIPPV to avoid endotracheal intubation in immunocompromised patients is appealing because, by assisting ventilation without having to invade the airway, it reduces infectious and hemorrhagic complications. Encouraging results derived from an uncontrolled series that reported NIPPV success rates as high as 67% (in forty-eight patients with AIDS and *Pneumocystis carinii* pneumonia).[140] Conti et al[141] avoided intubation in fifteen of sixteen patients treated with NIPPV with acute respiratory failure complicating hematologic malignancies, although patients were excluded if they had more than two organ-system failures or were responding poorly to antineoplastic therapy. More recently, Antonelli et al[142] randomized forty patients with acute respiratory failure of various etiologies following solid-organ transplantation to receive NIPPV or standard therapy. NIPPV reduced the need for intubation and lowered ICU mortality rate (both 20% vs. 50% in controls, $p < 0.05$), but total hospital mortality was similar. Trends for fewer health care–associated pneumonias and episodes of severe sepsis also were apparent among NIPPV-treated patients. In a subsequent study of fifty-two immunocompromised patients with respiratory failure, 58% with hematologic malignancies, randomized to receive NIPPV for at least 2 hours three times daily or standard $O_2$ therapy, those treated with NIPPV had fewer intubations (46% vs. 77%) and mortalities (50% vs. 81%, both $p < 0.05$).[143]

The sizable reductions in mortality among these high-risk patients strongly supports the use of NIV as the ventilatory modality of first choice in selected immunocompromised patients with acute respiratory failure. Patients developing respiratory insufficiency should be started on NIPPV relatively early,[133] before progression to severe respiratory failure, watched closely, and intubated without delay if needed. The need to perform invasive diagnostic procedures sometimes requires intubation in these patients although NIPPV can be used to support patients during fiber-optic bronchoscopy (see below).

### Acute Respiratory Distress Syndrome

The use of NIPPV to treat ARDS has been controversial, some considering it the "last frontier." One early cohort series reported that NIPPV maintained adequate oxygenation and averted intubation in six of twelve episodes of ARDS in ten patients.[144] In an observational cohort of seventy-nine patients with acute lung injury, Rana et al[145] found that shock, metabolic acidosis, and profound hypoxemia were predictors of NIV failure and cautioned against use in such patients. A more recent study used NIPPV as a "first-line" therapy of ARDS.[146] Of 479 patients presenting with ARDS, 147 had not been intubated upon admission to the ICU. These were begun on NIPPV and 54% avoided intubation. Not unexpectedly, outcomes were much better in those who avoided intubation than in those who failed NIPPV and required intubation: 2% versus 20% rate of ventilator-associated pneumonia and 6% versus 53% mortality rate, respectively. A simplified acute physiology score of equal to or less than 34 at baseline and a $Pa_{O_2}/FI_{O_2}$ greater than 175 after the first hour of NIPPV predicted success. Although not controlled, the trial suggests that a small minority (15% in this trial) of patients with ARDS can be managed successfully with NIPPV, but they must be chosen carefully and monitored very closely in an ICU. If their oxygenation fails to improve substantially within the first hour, urgent intubation should be considered.

Recently, a randomized controlled trial of NIPPV in patients with acute lung injury ($200 > Pa_{O_2}/FI_{O_2} < 300$ showed reductions in the need for intubation one of twenty on NIPPV pts and seven of 19 control patients) and actual intubations (one of twenty-one versus four of nineteen) as well as occurrence of organ failure (three of twenty-one versus fourteen of nineteen) (all $p < 0.05$), with a trend toward reduced mortality.[147] These studies suggest that NIPPV may have a role in the management of some patients with acute lung injury and ARDS, but it should be avoided in those with multiorgan system failure and very severe oxygenation defects who are likely to require prolonged ventilator support using sophisticated modes. If a trial of NIPPV is initiated, patients should be intubated without undue delay if they deteriorate or even fail to improve sufficiently.

### Trauma

Traumatic chest wall injuries such as flail chest or mild acute lung injury might respond favorably to NIPPV, but other etiologies might not. In a retrospective survey on forty-six trauma patients with respiratory insufficiency treated with NIPPV, Beltrame et al[148] found rapid improvements in gas exchange and a 72% success rate, but burn patients responded poorly. More recently, a controlled trial that randomized thoracic trauma patients with $Pa_{O_2}/FI_{O_2} < 200$ to NIPPV or high-flow oxygen was stopped early after enrollment of fifty patients because of significant reductions in intubation rate (12% vs. 40%) and hospital length of stay (14 vs. 21 days) in the NIPPV group.[149] These results support the use of NIPPV for hypoxemic respiratory failure in postthoracic trauma cases, but it is well to remember that these were carefully selected patients.

## Noninvasive Positive-Pressure Ventilation for Categories of Patients with Acute Respiratory Failure

## Do-Not-Intubate Patients

Some argue that there is little to lose by using NIPPV in almost any terminal patient. NIPPV could be used to lessen dyspnea, preserve patient autonomy, and permit verbal communication with loved ones during a terminal patient's final hours.[150] Some patients might be salvaged in the near term who otherwise would die without ventilatory assistance. This application is controversial, however, with some arguing that it could merely prolong the dying process, diminish patients' comfort in their waning hours, and promote excessive resource utilization.[151]

Among reports on NIPPV to treat patients who have declined or are reluctant to undergo intubation, Benhamou et al[152] retrospectively studied thirty such patients, mostly elderly men (mean age: 76 years) with COPD. Despite severe respiratory failure (mean $P_{A_{O_2}}$ of 43 mm Hg and $Pa_{CO_2}$ of 75 mm Hg), NIPPV was successful initially in 60% of patients. The authors considered NIPPV to be preferable to endotracheal intubation because short-term prognosis was better, and the modality appeared to be more comfortable with fewer complications. In another uncontrolled series, Meduri et al[153] observed a similar response to NIPPV among twenty-six patients with acute hypercapnic and hypoxemic respiratory failure who refused intubation.

In a prospective survey of 113 do-not-intubate patients treated with NIV,[154] amounting to 10% of all patients treated with NIPPV, survival to hospital discharge was 72% and 52% for patients with acute pulmonary edema and COPD, respectively, whereas it was less than 25% for patients with pneumonia or cancer. In addition, the absence of an effective cough and the inability to be awakened were significantly associated with hospital mortality. Similar findings were reported by Schettino et al.[15] Thus, the use of NIV may be justifiable in do-not-intubate patients who have a high likelihood of surviving the hospitalization. Longer-term survival of these hospital survivors, however, is poor; Chu et al[156] found a 30% 1-year survival for do-not-intubate patients with COPD as compared with 65% for patients desiring intubation. Also, no studies have yet assessed effects on patient comfort or family satisfaction.

NIV can also be used for palliation in patients whose prognosis is poor for surviving an admission to hospital, with the possible aims of alleviating dyspnea or prolonging survival long enough to enable a patient time to settle affairs or say goodbye to loved ones. As recommended by a consensus statement by a task force of the Society of Critical Care Medicine on NIV,[157] it is necessary for the patient, family, and caregivers to agree on these goals and to cease NIPPV promptly if it seems to be adding to suffering (via mask discomfort, for example) rather than alleviating it.

## Postoperative Patients

Several early case series on the use of NIPPV to treat respiratory insufficiency in postoperative patients with $Pa_{CO_2}$ values of greater than 50 mm Hg, $P_{A_{O_2}}$ values of less than 60 mm Hg, or evidence of respiratory muscle fatigue reported prompt reductions in respiratory rate and dyspnea scores, improvement in gas exchange, and high success rates (roughly 75%) in avoiding the need for reintubation.[158,159] Subsequent studies found that NIPPV was more effective than CPAP or chest physiotherapy in improving lung mechanics and oxygenation after coronary artery bypass surgery,[160] and better than $O_2$ therapy alone in improving oxygenation after lung-resection surgery.[161] NIPPV also ameliorated postgastroplasty pulmonary dysfunction in morbidly obese patients.[162]

These earlier studies heightened interest in the prophylactic use of CPAP or NIPPV after high-risk surgeries such as major abdominal surgery[162–165] or thoracoabdominal aneurysm repair.[166] These studies, using CPAP (10 cm $H_2O$) for 24 hours postoperation, observed reductions in the incidence of hypoxemia, pneumonia, atelectasis, and intubations compared with standard treatment.

Few studies have examined the postoperative role of NIPPV in patients with frank respiratory failure, but a randomized trial of NIPPV in forty-eight post–lung-resection patients with acute respiratory insufficiency, most with COPD, showed significant improvements in oxygenation and reductions in the need for intubation (21% vs. 50%) and mortality rate (13% vs. 38%) compared with conventionally treated controls (both $p < 0.05$).[167]

These studies strongly support the idea that both CPAP and NIPPV should be considered to prevent and treat postoperative respiratory complications and failure, especially in high-risk surgeries, but only a few studies have examined each of the various surgeries and positive-pressure techniques possible, so more specific recommendations cannot currently be made.

## Facilitation of Weaning

Nava et al[168] tested the hypothesis that NIPPV could be used to shorten the duration of invasive ventilation and reduce the occurrence of associated complications in a randomized, controlled trial of fifty patients intubated for acute respiratory failure secondary to COPD. If they failed a T-piece weaning trial performed 48 hours after intubation, patients were randomized to extubation followed by face-mask PSV or continued intubation and routine weaning. The NIPPV patients had higher overall weaning rates (88% vs. 68%), shorter durations of mechanical ventilation (10.2 vs. 16.6 days), briefer stays in the ICU (15.1 vs. 24 days), and improved 60-day survival rates (92% vs. 72%) (NIPPV-treated versus controls, all $p < 0.05$). In addition, no NIPPV-treated patients had nosocomial pneumonia compared with seven pneumonias among the controls. In a similar trial, Girault et al[169] randomized thirty-three patients with acute-on-chronic respiratory failure to remain intubated or to be extubated to NIPPV after failure of a 2-hour T-piece trial. The NIPPV group had a shorter duration of endotracheal intubation (4.6 vs. 7.7 days, $p = 0.004$), but the total duration of mechanical ventilation was longer in the NIPPV group, and weaning and mortality rates and ICU and hospital lengths of stay were similar between the groups.

More recently, Ferrer et al[170] randomized forty-three patients with "persistent" weaning failure (three consecutive failed T-piece trials) to be extubated to NIV or to remain intubated and be weaned using conventional methods. Patients randomized to NIV had shorter periods of intubation (9.5 vs. 20.1 days), shorter ICU (14 vs. 25 days) and hospital stays (14.6 vs. 40.8 days), a lower rate of nosocomial pneumonia (24% vs. 59%), and improved ICU and 90-day survivals (roughly 80% vs. 50%, all $p < 0.05$). This study lends strong support to the use of NIV to facilitate extubation, but it is worth noting that two-thirds of the patients had COPD or congestive heart failure.

In the most recent randomized (VENISE) trial that included 208 patients who had failed a spontaneous breathing trial, 69% of whom had COPD, the investigators included not only invasively ventilated and NIV groups, but also a group extubated early to oxygen therapy alone.[171] There were no differences in the main outcome variable: reintubation within 7 days (around one-third of patients in each group), or in most secondary outcome variables including mortality and lengths of stay in the ICU and hospital, complications and need for tracheostomy. Duration of intubation, however, was 1.5 days longer in the invasively ventilated group by design and the rate of postextubation acute respiratory failure was significantly less in the NIV group because NIV rescue was used after extubation in 57% and 45% of the oxygen therapy and invasively ventilated patients ($p > 0.05$), respectively, with an overall NIV rescue success rate of 52%. The authors concluded that, based on the reduction in postextubation acute respiratory failure, their results should "support the implementation of NIV" in difficult-to-wean patients, but recommended further study.

Thus, overall, the evidence favors the use of NIPPV to facilitate weaning and extubation in difficult-to-wean patients with COPD, mainly to reduce the occurrence of postextubation respiratory failure. In the absence of overwhelmingly favorable evidence, however, the following caveats should be borne in mind: (a) This approach should be reserved mainly for patients with COPD; (b) patients should be selected carefully, ascertaining that they are good candidates for NIPPV (see "Selection Guidelines" below); (c) patients should not have been difficult intubations; and (d) patients should be comfortable on levels of PSV that can used via mask after extubation.

## Treatment of Extubation Failure

Another potential application of NIPPV in the weaning process is to avoid reintubation in patients who fail extubation. Epstein et al[172] reported that extubation failure is associated with much higher morbidity and mortality rates (43%) than successful extubations (approximately 10%). Some investigators have used NIPPV prophylactically to see if extubation failure can be avoided. Jiang at al[173] randomized consecutive extubated patients to receive NIPPV or conventional therapy and found a trend for a higher reintubation rate in the NIPPV group (28% vs. 15%), suggesting that indiscriminate use of NIPPV is not effective for preventing extubation failure. Other investigators have attempted to prevent extubation failure by initiating NIPPV when patients develop risk factors for extubation failure. Esteban et al[174] tried this approach in a multicenter, multinational randomized trial of 221 patients developing risk factors for respiratory failure within 48 hours after they were extubated, including hypercapnia, tachypnea, or hypoxemia. Reintubation rates (48%) and ICU lengths of stay (18 days) were identical in both groups, and the study was terminated prematurely because of a significantly increased ICU mortality in the NIV group (25% vs. 15%, $p = 0.048$). The mortality difference was attributable to a higher mortality in the reintubated NIV patients, reintubation occurring almost 10 hours later than in the standard-therapy group. The authors concluded that NIV is not effective in unselected patients at risk for extubation failure and speculated that the greater delay in reintubation was responsible for the higher mortality rate. It is worth noting, however, that twenty-eight patients in the control group crossed over to NIPPV when they met failure criteria. Thus, the controls likely would have had a substantially higher reintubation rate had they not crossed over to NIPPV. Also, only 10% of patients enrolled in the study had COPD.

Another approach to treating extubation failure is to await the development of overt respiratory failure before initiating NIPPV. Hilbert et al[175] found that NIPPV used in this fashion lowered reintubation rate (20% vs. 67%) and shortened ICU lengths of stay in thirty patients with COPD and postextubation hypercapnic respiratory insufficiency compared with thirty historically matched controls. Keenan et al[176] randomized eighty-one patients to receive NIPPV or conventional therapy if they developed respiratory failure within 48 hours of extubation. The reintubation rate in this trial was roughly 70% in both the NIPPV group and controls, and no significant differences were found in hospital length of stay or survival. Patients with COPD, however, were excluded after the first year for ethical reasons, and only 12% of the patients had COPD. Furthermore, the pressures used (10 cm $H_2O$ inspiratory and 5 cm $H_2O$ expiratory) may have been insufficient to provide adequate ventilatory assistance.

Two subsequent randomized trials (consisting of ninety-seven and 162 patients, respectively,[177,178] approximately 30% to 40% of whom had COPD or congestive heart failure [CHF]) enrolled patients deemed to be at “high risk” for extubation failure. Both studies found that NIV reduced the need for reintubation, ICU mortality, and hospital length of stay, but hospital mortality was decreased only in the hypercapnic subgroup of the second study.[178] A more recent randomized trial of 106 patients with postextubation hypercapnia ($Pa_{CO_2}$ > 45 mm Hg) showed a significant reduction in postextubation respiratory failure as well as 90-day mortality in the group randomized to NIPPV compared to oxygen-treated controls.[179] The main effect of NIPPV in this study was to prevent greater retention of $CO_2$. The reason for the reduced 90-day mortality when mortality was not significantly reduced at earlier time points was not apparent.

These more recent studies support the use of NIV for patients who are at “high risk” for extubation failure, particularly if they have COPD, CHF, and/or hypercapnia. A Cochrane systematic review concluded that for patients who mainly have COPD, the evidence “demonstrated a consistent, positive effect on mortality and ventilator-associated pneumonia.”[180] Based on the Esteban study,[174] however, NIPPV to prevent extubation failure should be used very cautiously in at-risk patients who do not have COPD or other favorable characteristics because of the higher risk of NIPPV failure and its attendant morbidity and mortality. Such patients failing to improve promptly with NIPPV should be reintubated without delay.

## Pediatric Applications

Use of NIPPV in the pediatric population has been increasingly reported in the medical literature,[181] but most studies are retrospective and very few randomized controlled trials have been performed. For more detailed assessments of this literature, the reader is referred elsewhere.[181,182] Pediatric applications of NIPPV parallel those in the adult, but the etiologies of respiratory failure, of course, reflect those encountered in children. In earlier reports, Fortenberry et al[183] found that respiratory rate, $Pa_{CO_2}$, and oxygenation improved promptly after initiation of nasal bilevel NIPPV in a retrospective series of twenty-eight children with acute hypoxemic respiratory failure, only three of whom required intubation. Padman et al[184] subsequently reported similar results in a prospective series of thirty-four pediatric patients with both hypoxemic and hypoventilatory respiratory insufficiency, again with only three patients requiring intubation. More recent applications have been reported in children with status asthmaticus[185] and atelectasis[186] and randomized, controlled trials have been performed in children with bronchiolitis[187] and tracheomalacia.[188] These studies consistently show more rapid improvements in vital signs and gas exchange with NIPPV compared to controls.

These favorable responses to NIPPV in children are not surprising, particularly when they are suffering from conditions reported to be treated successfully by NIPPV in adults. Concerns have been raised, however, about treating very young children and infants with NIPPV because of increased nasal resistance[189] and inability to cooperate.[182] In the series of Fortenberry et al[183] of twenty-eight patients, NIPPV was successful in avoiding intubation in the ten children who were younger than 5 years of age. More recently, a retrospective trial observed successful application of NIPPV in infants following liver transplantation.[190] In a randomized, prospective trial on infants (median age: 9.5 months) with various causes of upper airway obstruction, CPAP and BiPAP proved to be equally efficacious in reducing respiratory rate and breathing effort; both were well tolerated, although BiPAP was associated with more asynchrony.[188] The lack of controlled trials makes it difficult to formulate specific guidelines for NIPPV in children, although reported success rates appear to be comparable to those in adults, and tentative guidelines have been proposed[182] that are based on those used for adults.

# Noninvasive Ventilation for Patients Undergoing Procedures

## Bronchoscopy and Endoscopy

NIPPV administered via T connector attached to a face mask can be used to assist ventilation and enhance oxygenation during fiber-optic bronchoscopy in high-risk patients. First reported to improve oxygenation and be well tolerated in a small series of immunocompromised hypoxemic patients, NIPPV was shown subsequently to improve and sustain oxygenation better than conventional $O_2$ supplementation in a randomized, prospective trial in twenty-six patients with $Pa_{O_2}/FI_{O_2}$ ratios of 200 or less and suspected nosocomial pneumonia.[191] More recently, NIPPV to assist ventilation during fiber-optic bronchoscopy was administered successfully using a “helmet” device (see techniques section below).[192] NIPPV also has been used to assist ventilation during upper endoscopy for gastric tube placement in patients with neuromuscular disease[193] and during performance of transesophageal echocardiography.[194]

## Preoxygenation before Intubation

NIPPV can be used to improve $O_2$ saturation during and after intubation and decrease the incidence of $O_2$ desaturations below 80% during intubation according to a randomized trial in critically ill patients with hypoxemic respiratory failure.[195] This approach is promising but needs further evaluation before routine use can be recommended. This also begs the question whether, if NIV improves oxygenation substantially, intubation could be avoided in some of these patients.

# Selection of Patients for Noninvasive Positive-Pressure Ventilation in the Acute Care Setting

## Determinants of Success

Retrospective[133,196] and prospective[197–199] studies have identified predictors of NIPPV success (Table 18-2). Patients with baseline hypercapnia fare better than those with hypoxemia alone, but successful patients have lower baseline $Pa_{CO_2}$ values (79 vs.

98 mm Hg) and higher pH values (7.28 vs. 7.22) than failure patients.[105] Pneumonia predisposes to failure whether alone or in combination with COPD (odds ratio [OR] 5.63 for COPD).[197,199] The strongest predictor of success, though, is prompt improvement in gas exchange and heart and respiratory rates within 1 to 2 hours of NIPPV initiation.[133,196–199] Confalonieri et al[198] have incorporated some of these predictors (Acute Physiology and Chronic Health Evaluation [APACHE] II score ≥ 29, pH < 7.25, Glasgow Coma Score ≤ 11, and respiratory rate ≥ 35 breaths/min) into two risk charts for NIPPV failure, one to be used at baseline and the other at 2 hours (Fig. 18-2). If these abnormalities were all present at baseline, the likelihood of failure was 82% and rose to 99% if they persisted at 2 hours.

**Table 18-2: Predictors of Success during Acute Applications of Noninvasive Positive-Pressure Ventilation**

| |
|---|
| Younger age |
| Lower activity of illness (APACHE score) |
| Able to cooperate, better neurologic score |
| Able to coordinate breathing with ventilator |
| Less air leaking, intact dentition |
| Tachypnea, but not excessively rapid (> 24 but < 35 breaths/min) |
| Hypercarbia, but not too severe ($Pa_{CO_2}$ > 45 but < 92 mm Hg) |
| Acidemia, but not too severe (pH < 7.35 but > 7.10) |
| Improvements in gas exchange, heart and respiratory rates within first 2 hours* |

*Most powerful predictor.

*Source*: Adapted, with permission, from Ambrosino et al,[133] Soo Hoo et al,[196] and Antonelli et al.[199]

**Figure 18-2**

**Admission**

| | | pH admission < 7.25 | | pH admission 7.25–7.29 | | pH admission > 7.30 | |
|---|---|---|---|---|---|---|---|
| | RR | Apache ≥ 29 | Apache < 29 | Apache ≥ 29 | Apache < 29 | Apache ≥ 29 | Apache < 29 |
| GCS 15 | <30 | 29 | 11 | 18 | 6 | 17 | 6 |
| | 30–34 | 42 | 18 | 29 | 11 | 27 | 10 |
| | ≥35 | 52 | 24 | 37 | 15 | 35 | 14 |
| GCS 12–14 | <30 | 48 | 22 | 33 | 13 | 32 | 12 |
| | 30–34 | 63 | 34 | 48 | 22 | 46 | 21 |
| | ≥35 | 71 | 42 | 57 | 29 | 55 | 27 |
| GCS ≤11 | <30 | 64 | 35 | 49 | 23 | 47 | 21 |
| | 30–34 | 76 | 49 | 64 | 35 | 62 | 33 |
| | ≥35 | 82 | 59 | 72 | 44 | 70 | 42 |

**1 Hour**

| | | pH after 2 h < 7.25 | | pH after 2 h 7.25–7.29 | | pH after 2 h ≥ 7.30 | |
|---|---|---|---|---|---|---|---|
| | RR | Apache ≥ 29 | Apache < 29 | Apache ≥ 29 | Apache < 29 | Apache ≥ 29 | Apache < 29 |
| GCS 15 | <30 | 72 | 35 | 27 | 7 | 11 | 3 |
| | 30–34 | 88 | 59 | 49 | 17 | 25 | 7 |
| | ≥35 | 93 | 73 | 64 | 27 | 38 | 11 |
| GCS 12–14 | <30 | 84 | 51 | 41 | 13 | 19 | 5 |
| | 30–34 | 93 | 74 | 65 | 28 | 39 | 12 |
| | ≥35 | 96 | 84 | 78 | 42 | 54 | 20 |
| GCS ≤11 | <30 | 93 | 74 | 65 | 28 | 39 | 12 |
| | 30–34 | 97 | 88 | 83 | 51 | 63 | 26 |
| | ≥35 | 99 | 93 | 90 | 66 | 76 | 40 |

Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com


Charts showing percentage likelihood of noninvasive positive-pressure ventilation (*NIPPV*) failure in patients with chronic obstructive pulmonary disease (*COPD*) based on pH and Acute Physiology and Chronic Health Evaluation (*APACHE*) score and Glasgow coma score (*GCS*). Chart on top refers to values at the time of admission, and bottom chart is at 2 hours. Numbers in boxes are percentages for likelihood of failure. At 2 hours, the combination of an APACHE score of less than 11, APACHE score of 29 or greater, and a Glasgow coma score of less than 11 predicts failure with a likelihood of 99%. Based on 1033 patients with COPD treated with NIPPV in thirteen different experienced units. (Used, with permission, from Antonelli, et al. Predictors of failure of noninvasive positive pressure ventilation in patients with acute hypoxemic respiratory failure: a multi-center study. *Intensive Care Med.* 2001;27:1718–1728. With kind permission from Springer Science and Business Media.)

Timing of initiation is another determinant of success. Ambrosino et al[133] advised that NIPPV "should be instituted early in every patient before a severe acidosis ensues." Initiation of NIPPV should be viewed as taking advantage of a "window of opportunity." The window opens when acute respiratory distress occurs and shuts when the patient deteriorates to the point of necessitating immediate intubation. In this context, it should be emphasized that NIPPV is used as a way of preventing intubation, not replacing it.

## Selection Guidelines

The preceding predictors of success and failure and the entry criteria used for enrollment of patients into the many studies have served as a basis for consensus guidelines on the selection of patients to receive NIPPV for acute respiratory failure.[200] These guidelines use a simple three-step approach outlined in Table 18-3. The first step asks whether the patient needs ventilator assistance. Patients with mild or no respiratory distress are excluded from consideration because they should do well without ventilator assistance. Those needing ventilator assistance are identified using clinical indicators of acute respiratory distress and gas-exchange derangement, as listed in Table 18-3. These criteria are most applicable to patients with COPD but can be used to screen those with other forms of expiratory failure, although some modifications are advisable. For example, studies on NIPPV in acute pulmonary edema and acute hypoxemic respiratory failure have used higher respiratory rates as enrollment criteria (> 30 to 35 instead of > 24 breaths/min) and a $P_{CO_2}/FI_{O_2}$ ratio of less than 200.[111–114]

**Table 18-3: Selection Guidelines: Noninvasive Ventilation for Patients with Acute Respiratory Failure**

Step 1. Identify patients with reversible causes for acute respiratory failure (ARF) in need of ventilator assistance

a. Reversible cause for ARF such as COPD or acute cardiogenic pulmonary edema

b. Symptoms and signs of acute respiratory distress:

i. Moderate to severe dyspnea, increased over usual

ii. Rate > 24 for COPD, rate > 30 for hypoxemic ARF

iii. Accessory muscle use, paradoxical breathing

c. Gas-exchange abnormalities:

i. $Pa_{CO_2}$ > 45 mm Hg, pH < 7.35

ii. $Pa_{O_2}/FI_{O_2}$ < 200

Step 2. Exclude those at increased risk with noninvasive ventilation:

a. Respiratory arrest

b. Medically unstable (hypotensive shock, uncontrolled cardiac ischemia or arrhythmias)

c. Unable to protect airway (impaired cough or swallowing mechanism)

d. Excessive secretions

e. Agitated or uncooperative

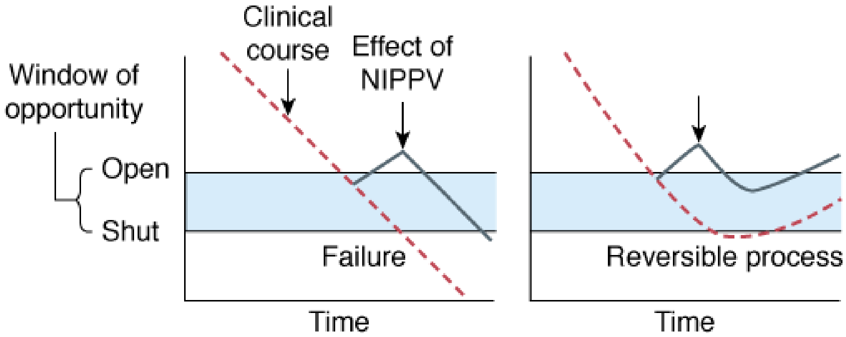
f. Facial trauma or burns or anatomic abnormalities interfering with mask fit

*Source:* Reprinted with permission of the American Thoracic Society. Copyright © 2012 American Thoracic Society. From Mehta S, Hill NS. Noninvasive ventilation: state of the art. *Am J Respir Crit Care Med.* 2001;163:540–577. Official Journal of the American Thoracic Society.

The second step is to screen out patients in whom use of NIPPV is contraindicated (see Table 18-3). Most are relative contraindications, and judgment should be exercised in implementing them. Also, some conditions that have been listed as contraindications in the past no longer preclude the use of NIPPV. For example, patients with coma, if related to hypercapnia, may be managed successfully with NIPPV. Diaz et al[201] demonstrated that patients (mainly with COPD) with hypercapnic coma (Glasgow coma scores ≤ 8) had outcomes just as good as those with higher Glasgow coma scores. In another report,[152] nearly half the patients were obtunded or somnolent initially, yet most were managed successfully with NIPPV.

The underlying etiology and potential reversibility of acute respiratory failure are also important considerations in patient selection. As discussed previously, the strongest evidence supports the use of NIPPV for COPD and either NIPPV or CPAP for acute cardiogenic pulmonary edema. As illustrated in Figure 18-3, a reversible etiology permits the use of NIPPV as a “crutch” that assists the patient through a critical interval, allowing time for other therapies such as bronchodilators, steroids, or diuretics to reverse the underlying condition. More severe, less easily reversed forms of respiratory failure that will require prolonged periods of ventilatory support, such as status asthmaticus requiring controlled hypoventilation, complicated pneumonias, or ARDS, are best managed using invasive ventilation. Marginal patients might be considered for a trial of NIPPV but should be watched closely for further deterioration so that needed intubation is not delayed unduly.

**Figure 18-3**




Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com


Schema to illustrate importance of reversibility in success of noninvasive positive-pressure ventilation (*NIPPV*) for acute respiratory failure. *Left:* Relentless downhill clinical course of underlying process (severe acute respiratory distress syndrome). NIPPV initiated when window of opportunity opens still fails because it is not sufficient to support the patient when the underlying process has progressed too far. *Right:* NIPPV succeeds when medical therapy reverses the underlying process (chronic obstructive pulmonary disease exacerbation).

## Noninvasive Positive-Pressure Ventilation for Chronic Respiratory Failure

### Evidence for Efficacy

#### Restrictive Thoracic Diseases

Restrictive thoracic disorders limit expansion of the thoracic cage either because of increased elastance (as with chest wall deformity), muscle weakness (as with neuromuscular diseases), or both. Experience gained during the polio epidemics of the 1930s through 1950s and subsequently with other neuromuscular disorders demonstrated that "body ventilators," even when used intermittently, were effective at stabilizing or even reversing chronic respiratory failure secondary to restrictive thoracic disorders.[202,203] Although NIPPV had been available for the therapy of restrictive thoracic diseases for decades,[204] its use beyond a few centers with special expertise was quite limited until the late 1980s. Mouthpieces or full-face masks designed mainly for administration of anesthesia were the only interfaces available, posing challenges for patient adaptation. Despite this, one center reported remarkable success with mouthpiece ventilation in a cohort of 257 patients with neuromuscular disease and chronic respiratory failure that was treated for an average of 9.6 years.[205] Bulbar function was intact, including speech and swallowing, but the patients were otherwise severely compromised. Of these patients, 144 required 20 to 24 hours of ventilator support daily and had vital capacities of only 10.5% of predicted. Nonetheless, sixty-seven were switched successfully from tracheostomies to NIPPV, and only thirty-eight died during the follow-up interval of up to 37 years. Although the study was uncontrolled, the authors concluded that mouthpiece NIPPV prolonged survival and enhanced convenience and communication in these severely compromised patients.

Despite this favorable experience, however, wider use of NIPPV awaited further developments during the early 1980s. One was the development of nasal CPAP for therapy of obstructive sleep apnea that encouraged the creation of more comfortable commercially available nasal masks.[206] Another was the suggestion by French investigators that nasal ventilation could be used in patients with muscular dystrophy to halt progression of the disease.[23] In 1987, several small series and case reports appeared describing successful management with nocturnal nasal ventilation of patients with chronic respiratory failure secondary to a variety of restrictive thoracic disorders.[207–208] Subsequently, additional small series have confirmed the earlier findings, but no prospective, randomized series have been performed, mainly for ethical reasons. Despite this, NIPPV has gained wide acceptance as the modality of first choice to treat patients with chronic respiratory failure.[200] Because most series have combined patients with different etiologies, the following will discuss evidence for benefits attributed to NIPPV for restrictive thoracic diseases in general, making reference to individual diagnoses as appropriate. Table 18-4 lists individual diagnoses of neuromuscular diseases reported to benefit from NIPPV.

**Table 18-4: Restrictive Thoracic Diseases Treated with Noninvasive Ventilation**

| |
|---|
| **Recommended for the following diagnoses**:<br>Chest wall deformity<br>Kyphoscoliosis<br>Postthoracoplasty for tuberculosis<br>Slowly progressive neuromuscular disorders<br>Postpolio syndrome<br>High spinal cord injury<br>Spinal muscular atrophy<br>Slowly progressive muscular dystrophies<br>Duchenne muscular dystrophy<br>Limb-girdle muscular dystrophy<br>Myotonic dystrophy<br>Multiple sclerosis<br>Bilateral diaphragmatic paralysis<br>More rapidly progressive neuromuscular disorders[*]<br>Amyotrophic lateral sclerosis |
| **Not recommended for[†]:**<br>Rapidly progressive neuromuscular disorders<br>Guillain-Barré syndrome<br>Myasthenia gravis |

[*]Tracheostomy ventilation should be considered in far-advanced cases.

[†]Unless upper airway protective mechanisms intact.

### Effects on Symptoms and Daytime Gas Exchange

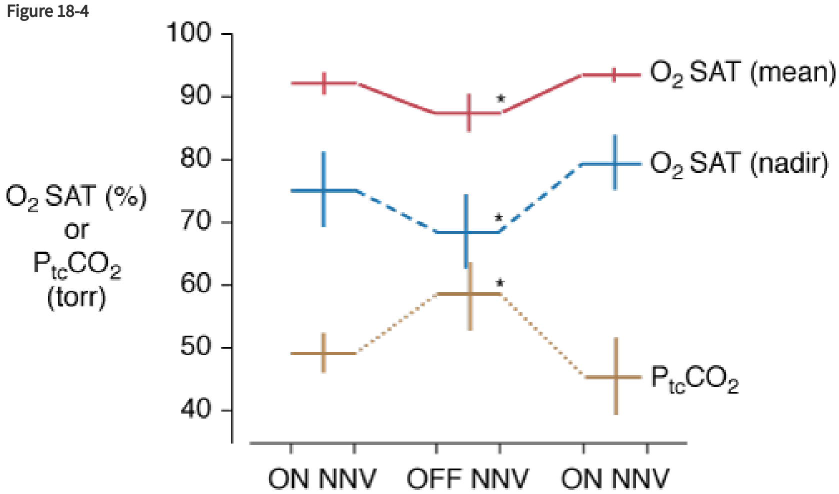
Numerous studies on the efficacy of NIPPV in a wide variety of neuromuscular and chest wall disorders have shown that intermittent NIPPV consistently improves symptoms of fatigue, daytime hypersomnolence, and morning headache.[210–213] Daytime gas exchange also improves, $Pa_{CO_2}$ dropping on average from 63 mm Hg to 48 mm Hg and $Pa_{O_2}$ increasing from 54 to 71 mm Hg among multiple studies.[1] The improvement in gas exchange is gradual, usually occurring over a period of weeks, as patients increase their hours of use, mainly at night. In addition, nasal NIPPV has been shown to ameliorate chronic hypoventilation in patients with severe kyphoscoliosis who fail to improve with nasal CPAP alone.[214]

### Effects on Nocturnal Gas Exchange and Sleep

Neuromuscular and chest wall disorders cause protean sleep-related breathing disturbances depending on the specific syndrome and the involvement of respiratory muscles.[215] Abnormalities include obstructive and central sleep apneas, intermittent desaturations, particularly during rapid eye movement sleep in patients with diaphragmatic weakness, and sustained hypoventilation as global weakness and/or chest wall restriction advances. These abnormalities often lead to sleep disruption, characterized by diminished sleep duration, fragmentation related to arousals, and poor sleep quality secondary to diminished slow-wave and rapid eye movement sleep.

NIPPV ameliorates nocturnal hypoventilation and has been shown to eliminate the intermittent obstructive apneas and severe $O_2$ desaturations that occur during negative-pressure ventilation, particularly during rapid eye movement sleep.[215] Although no randomized, prospective trials have been performed to investigate effects of NIPPV on sleep, investigators have evaluated efficacy using temporary withdrawal of nocturnal nasal NIPPV from long-term users with restrictive thoracic diseases whose gas exchange had been improved by prior NIPPV use.[216,217] Temporary withdrawal of NIPPV caused a deterioration of nocturnal oxygenation and ventilation[215] (Fig. 18-4) and increased frequency of arousals.[216] All changes were reversed promptly on resumption of NIPPV. These findings indicate that NIPPV is important in preventing deterioration of nocturnal gas exchange that is thought to predispose to chronic hypoventilation and contribute to frequent arousals and fragmented sleep. Stabilization of nocturnal gas exchange and improved sleep quality are thought to be major reasons for the improvement in symptoms associated with NIPPV use. Although sleep quality is improved compared to no ventilator assistance, air leaking during NIPPV, however, can contribute to sleep fragmentation.[218] Atkeson et al[219] found that asynchrony with the ventilator was very prevalent in patients with amyotrophic lateral sclerosis (ALS) using NIPPV nocturnally. Twenty-three patients had an average asynchrony index of 69/hour constituting 17% of sleep time.

Figure 18-4



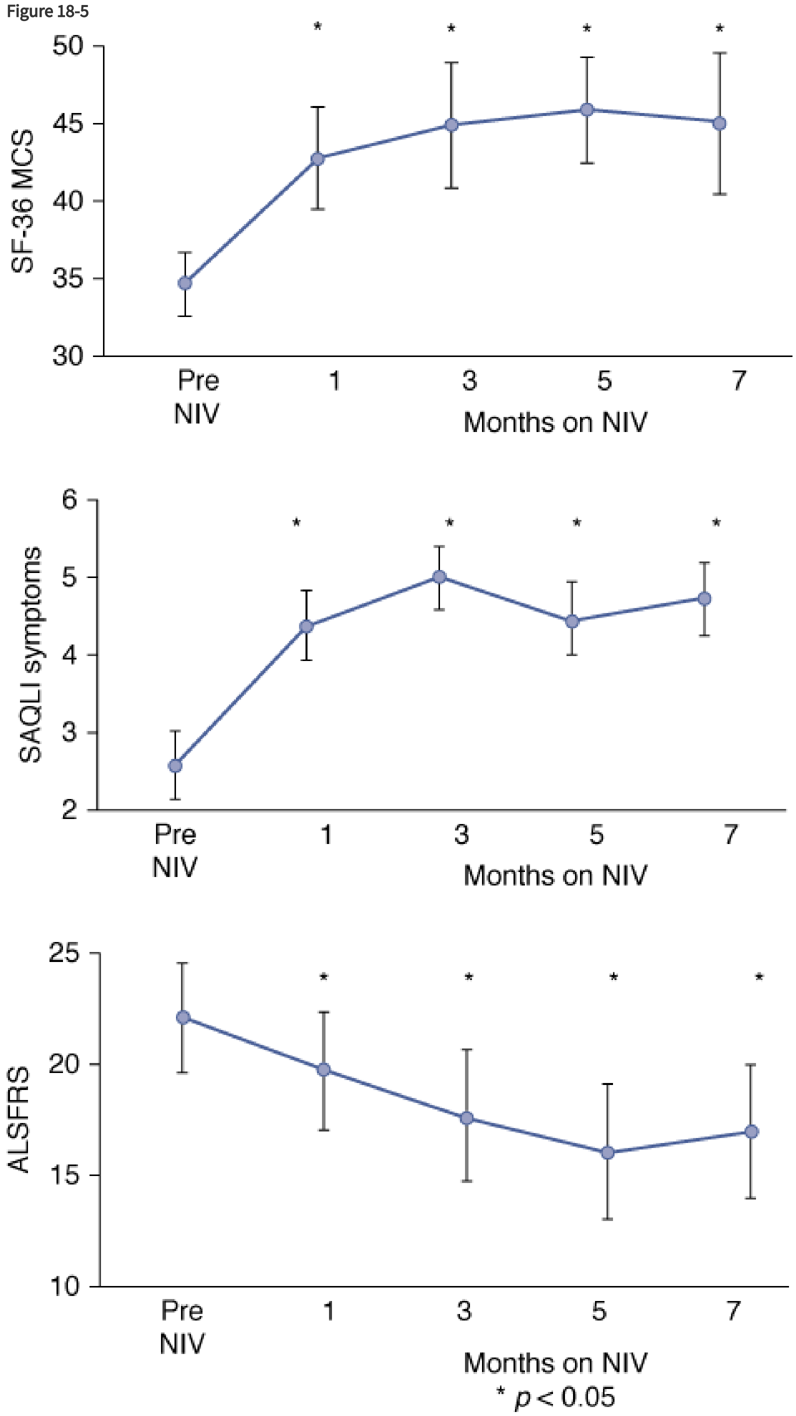



Effect of withdrawal of nocturnal nasal ventilation (*NNV*) on mean and nadir $O_2$ saturations and mean transcutaneous $P_{CO_2}$ levels obtained during nocturnal monitoring. Values on the left labeled “on NNV” were obtained during NNV use on the night before NNV withdrawal, values in the middle were obtained on the last night of the NNV withdrawal period, and values on the right labeled “on NNV” were obtained a week after NNV was resumed. Data are mean ± standard error (SE). Asterisk indicates $p < .05$ compared with “on NNV” values ($n = 6$). (Used, with permission, from Hill NS, Eveloff SE, Carlisle CC, Goff SG. Efficacy of nocturnal nasal ventilation in patients with restrictive thoracic disease. *Am Rev Respir Dis.* 1992;145:365–371.)

### Effects on Quality of Life

NIPPV improves health-related quality of life in patients with restrictive thoracic disorders.[220,221] The nature and duration of improvement, however, depend on the natural history of the underlying disorder. Among patients with ALS, the mental component summary score[222] and the vitality score[223] on the SF (short form)-36 questionnaire register sustained improvements. These improvements are seen even when the functional rating declines (Fig. 18-5), but attrition rates are high over time related to the high mortality of the underlying condition.[224] Bourke et al performed a randomized, controlled trial in forty-one patients with ALS, demonstrating improved quality of life in patients with normal or only mild to moderately impaired bulbar dysfunction, which was sustained for up to 2 years.[225] Even more sustained would be expected, of course, in conditions with lower rates of progression. Likewise, musculoskeletal functional gains would not be expected unless the underlying condition offers the potential for improvement (i.e., no quadriplegia).

Figure 18-5




Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com



Serial data for the Short Form-36 Mental Component Summary (*SF-36 MCS*), Sleep Apnea Quality of Life Index (*SAQLI*) symptoms domain, and ALS Functional Rating Scale (*ALSFRS*) immediately before starting noninvasive ventilation (*NIV*) and after 1, 3, 5, and 7 months of NIV. The slight improvement in the ALSFRS score at 7 months is secondary to a survivor effect. (Used, with permission, et al. A prospective study of quality of life in ALS patients treated with noninvasive ventilation. *Neurology.* 2001;57:153–156.)

Some studies have compared quality of life during use of NIPPV with that during use of tracheostomy positive-pressure ventilation among patients with restrictive thoracic diseases.[226–228] Both groups have high levels of satisfaction, but NIPPV was rated as preferable to ventilation via a tracheostomy with regard to comfort, convenience, portability, and overall acceptability.[226] Although tracheostomy ventilation received higher scores for quality of sleep and providing a sense of security, the vast majority of patients preferred NIPPV. Another survey of thirty-five ventilator users, with twenty-nine NIPPV users and six with tracheostomy ventilation, found satisfactory levels of psychosocial functioning and mental well-being, as determined by standard questionnaires.[227] The ratings compared favorably with those of a general population, and NIPPV and tracheostomy ventilation received similar scores. In another survey of home mechanical ventilation users, patients with scoliosis receiving tracheostomy ventilation had higher health-index ratings than those receiving NIPPV.[228] Thus, NIPPV offers many advantages over invasive mechanical ventilation, including comfort, portability, cost, and convenience, but some ratings, including overall health and sense of security, may favor invasive mechanical ventilation.

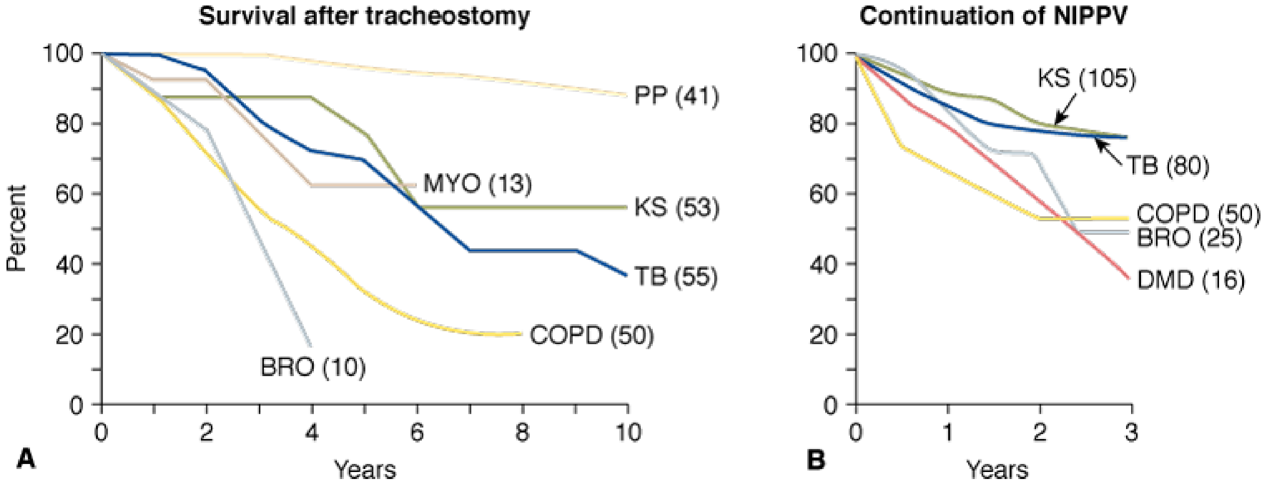
#### Effect on Hospital Utilization

In their long-term follow-up study of patients with kyphoscoliosis, sequelae of tuberculosis, and Duchenne muscular dystrophy, Leger et al[229] observed significant reductions in hospital days per patient per year from 34, 31, and 18 days for the year before starting NIPPV to 6, 10, and 7 days for the year after, respectively. These findings suggest that NIPPV may cut health care resource utilization in these patients, with the potential for substantial cost savings. More recently, Janssens et al[230] observed a similar reduction in hospital days per patient per year (median of 17 days for the year before and 6 days for the year after NIPPV) for patients with restrictive thoracic disorders in a long-term retrospective study from Switzerland. Although these uncontrolled studies do not preclude the possibility that changes in hospitalization practices over time could have been responsible for the reductions, it appears highly likely that NIPPV reduces the need for hospitalization among these patients.

#### Effect on Survival

Long-term cohort series provide strong evidence that NIPPV prolongs survival in comparison with unventilated patients. The ability to prolong survival is obvious in patients using NIPPV continuously who would die if ventilator assistance stops for more than a few minutes.[205] Studies from France[229] and England[231] report on several hundred patients with chronic respiratory failure of various etiologies treated with intermittent nasal NIPPV for periods of up to 5 years (Fig. 18-6). Rather than survival rates, these studies used the rate for continuation of NIPPV, thought to correspond closely with survival for most diagnoses. The studies found very favorable continuation rates for postpolio and kyphoscoliosis patients (approximately 100% and 80%, respectively, after 5 years). Patients with sequelae of old tuberculosis had higher continuation rates in the British study compared with the French study (94% vs. 60%, respectively), perhaps reflecting the greater morbidity of the French patients at the time of enrollment. Also, patients with Duchenne muscular dystrophy in the French study had lower continuation rates (47%) than did those with other neuromuscular diseases, with 28% undergoing tracheostomy and the remaining 25% dying. A more recent follow-up on the English cohort observed much better survival rates for patients with Duchenne muscular dystrophy treated with NIPPV than those previously reported by the French group: 85% for 1 year and 73% for 5 years.[232] This disparity may reflect differences in the severity of illness between patients when begun on NIPPV, how cough was assisted, or practices on switching to tracheostomy. Overall, the continuation rates from these long-term follow-up studies are similar to those observed for patients treated with invasive ventilation reported earlier by the French group.[233]

**Figure 18-6**




Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com


**A.** Survival after tracheostomy in 222 patients with chronic respiratory failure of various etiologies followed for as long as 10 years. (Redrawn, with permission, from Robert et al.[233]) **B.** Likelihood of continuing noninvasive positive-pressure ventilation (*NIPPV*) in 276 patients with chronic respiratory failure followed for as long as 3 years. (Redrawn, with permission, from Leger et al.[229]) *BRO,* bronchiectasis; *COPD,* chronic obstructive pulmonary disease; *DMD,* Duchenne muscular dystrophy; *KS,* kyphoscoliosis; *MYO,* myopathy; *PP,* postpolio syndrome; *TB,* history of tuberculosis.

Survival would be anticipated to be shorter in patients with ALS than in those with more slowly progressive neuromuscular diseases. In a prospective, nonrandomized trial on twenty consecutive patients, Pinto et al[234] treated the first ten with medical therapy alone and the next ten with NIPPV. After 2 years, 50% of the NIPPV patients were alive, whereas all the medical therapy patients had died. Aboussouan et al[235] compared the survival of thirty-one patients with ALS who continued NIPPV with that of twenty-one patients who were intolerant of NIPPV. The intolerant patients had a significantly greater risk of dying over the 3-year study period than were those who remained on NIPPV (relative risk: 3.1). As might be anticipated, patients with bulbar involvement were unlikely to be tolerant of NIPPV (only six of twenty); if they could tolerate the therapy, however, it imparted an apparent survival advantage. Bach[236] has reported that NIPPV prolongs survival and postpones the need for tracheostomy in ALS. In the randomized, controlled trial on ALS by Bourke et al, patients without severe bulbar involvement had an average 205-day prolongation of survival.[225] Those with severe bulbar involvement tolerated NIPPV much less well than did those without, and had no survival benefit, but they did have some improvement of sleep-related symptoms if they were tolerant.

Based on the above observations, NIPPV is now used in approximately 15% of patients with ALS compared with only 2% using invasive mechanical ventilation, according to one recent cross-sectional U.S. survey.[237] On the other hand, Bach[236] also found that after 5 years of follow-up, only eight of twenty-five (32%) patients with ALS using NIPPV were alive compared with twenty-seven of fifty (54%) patients receiving tracheostomy ventilation. This suggests that NIPPV is less effective at prolonging survival than tracheostomy ventilation in ALS, as might be anticipated among patients with a neuromuscular disease that impairs bulbar function.

## Selection of Patients with Restrictive Thoracic Disease to Receive Noninvasive Positive-Pressure Ventilation

Table 18-5 lists guidelines for the selection of patients with restrictive thoracic disorders to receive NIV based on American College of Chest Physicians consensus,[238] Canadian Thoracic Society clinical practice guideline,[239] and Medicare guidelines in the United States. Patients should have symptomatic nocturnal hypoventilation manifested by poor sleep quality, such as morning headache, daytime hypersomnolence, and low energy in combination with demonstrable daytime or sustained nocturnal hypoventilation. The duration of nocturnal $O_2$ desaturation used as an indicator of nocturnal hypoventilation (< 88% for more than 5 consecutive minutes) was suggested by consensus,[238] but has never been validated. Even if symptoms are minimal or lacking, patients with severe $CO_2$ retention (> 50 mm Hg) and those recovering from bouts of acute respiratory failure are considered for long-term NIV, particularly if there is persistent $CO_2$ retention or a history of repeated hospitalizations. The consensus group of the American College of Chest Physicians as well as the Canadian Thoracic Society also recommended NIPPV for patients with severe pulmonary dysfunction (FVC < 50% of predicted or maximal inspiratory pressure < 60 cm $H_2O$), even in the absence of $CO_2$ retention, despite the lack of evidence from clinical studies to support the initiation of NIV on the basis of pulmonary function alone. Simonds also suggests that other possible indications for NIV in patients with chronic respiratory insufficiency include infectious complications, pregnancy, and the perioperative state.[240]

**Table 18-5: Selection Guidelines: Long-Term Noninvasive Ventilation for Restrictive Thoracic Disorders or Obesity Hypoventilation**

| |
|---|
| **Indications**<br>1. Symptoms: fatigue, morning headache, hypersomnolence, nightmares, enuresis, dyspnea, and so on<br>2. Signs: cor pulmonale<br>3. Gas-exchange criteria:<br>Daytime $Pa_{CO_2}$ ≥ 45 mm Hg<br>Nocturnal oxygen desaturation ($Sa_{O_2}$ ≤ 88% for more than 5 minutes sustained or >10% of total monitoring time)<br>4. Sleep evaluation:<br>Unnecessary in restrictive thoracic disorders if other criteria met<br>Should be obtained for obesity hypoventilation to assess obstructive sleep apnea*<br>5. Other possible indications<br>Recovering from acute respiratory failure with persistent $CO_2$ retention<br>Repeated hospitalizations for acute respiratory failure |
| **Contraindications**<br>Inability to protect airway<br>Impaired cough<br>Impaired swallowing with chronic aspiration<br>Copious secretions<br>Need for continuous or nearly continuous ventilatory assistance<br>Anatomic abnormalities that interfere with mask fitting<br>Poorly motivated patient or family<br>Inability to cooperate or comprehend therapy<br>Inadequate financial or caregiver resources |

*If obstructive sleep apnea is present, trial of continuous positive airway pressure may be warranted (see Table 18-6).

*Source:* Reprinted with permission of the American Thoracic Society. Copyright © 2012 American Thoracic Society. From Mehta S, Hill NS. Noninvasive ventilation: state of the art. *Am J Respir Crit Care Med.* 2001;163:540–577. Official Journal of the American Thoracic Society.

Relative contraindications to the use of NIPPV for chronic respiratory failure (see Table 18-5) include inability to protect the upper airway because of impaired cough or swallowing or excessive secretions. Aggressive treatment with techniques or devices to assist cough[241] may permit the use of NIPPV in such patients who otherwise would not be candidates, but if the condition is too severe, tracheostomy is indicated if the patient desires maximal prolongation of life. Tracheostomy ventilation has been recommended when the need for ventilator assistance exceeds 16 hours daily,[242] although many patients still prefer NIPPV.[226] Table 18-5 lists other relative contraindications to NIPPV. The clinician must render a judgment as to whether these are sufficient to preclude a trial of NIV.

The natural history of the restrictive thoracic disorder also should be considered when deciding about NIPPV. Patients with chest wall deformities and stable or slowly progressive neuromuscular disorders respond well to NIPPV and remain stable for long periods of time.[229,231] Patients with more rapidly progressive neuromuscular disorders such as ALS may respond well temporarily, but as debility progresses and bulbar function deteriorates, NIPPV loses its efficacy. Those who wish to optimize their chances for survival may prefer invasive ventilation, and others may desire hospice care. Patients with rapidly progressive neuromuscular conditions such as Guillain-Barré syndrome or myasthenia gravis in crisis usually are poor candidates for NIV because swallowing frequently is impaired when ventilatory dysfunction becomes severe.

## When to Start Long-Term Noninvasive Positive-Pressure Ventilation for Restrictive Thoracic Disorders

NIPPV to treat chronic respiratory failure secondary to restrictive thoracic diseases has gained wide acceptance, but the optimal time for initiation has been debated. Prophylactic initiation in progressive neuromuscular diseases, before the onset of symptoms or daytime hypoventilation, has been proposed to retard the progression of respiratory dysfunction. Raphael et al[243] tested this hypothesis in seventy-six patients with Duchenne muscular dystrophy who had not yet developed symptoms or daytime hypoventilation, randomizing them to receive nasal NIPPV or standard therapy. Not only did NIPPV fail to slow disease progression, it also was associated with greater mortality, leading to premature termination of the trial. The authors surmised that mortality was increased because NIPPV gave patients a false sense of security that caused them to delay seeking medical attention when they developed respiratory infections. The study had numerous shortcomings, including failure to document patient adherence or to consistently use techniques to assist cough, but it has stemmed any enthusiasm about using prophylactic NIPPV in patients with Duchenne muscular dystrophy. Ward et al[244] randomized twenty-six patients with neuromuscular disease and nocturnal hypoventilation to start NIPPV right away or to await the onset of daytime hypercapnia before starting. Patients starting promptly had better gas exchange and quality of life, and a trend toward fewer hypercapnic crises compared to those starting later. The authors concluded that NIPPV is best started with the onset of symptomatic nocturnal hypoventilation and before the onset of diurnal hypercapnia.

Early initiation of NIPPV also has been proposed to treat patients with ALS.[245] Current guidelines based on expert consensus recommend starting NIPPV if FVC drops below 50% of predicted[246] or maximal inspiratory pressure drops below 60 cm $H_2O$. In a preliminary study,[247] twenty patients with ALS and FVC values ranging between 70% and 100% of predicted were randomized to receive NIPPV if they had an $SaO_2$ of less than 90% for more than 1 minute during nocturnal oximetry ("early intervention") or to await a drop in FVC to less than 50% of predicted ("standard of care"). The early intervention group had a significant increase in the vitality subscale on the SF-36, suggesting that earlier intervention might offer some benefit in patients with ALS, but more research is necessary.

Presently, awaiting the onset of symptoms of nocturnal hypoventilation before initiation of NIPPV is the most pragmatic approach because adherence to therapy is often poor unless patients are motivated by the desire for symptom relief. Symptomatic patients who have only nocturnal but no daytime hypoventilation, as demonstrated by frequent, sustained nocturnal $O_2$ desaturations, are good candidates for initiation. Masa et al[247] showed improvements in dyspnea scores, morning headache, and confusion after 2 weeks of nocturnal NIPPV in twenty-one such patients, whose proportion of sleep time with an $SaO_2$ of less than 90% averaged 40% to 50% on room air before initiation of NIPPV and fell to 6% afterward. The timing of initiation requires a judgment based on the anticipated progression of the disease (sooner for more rapid progression), the patient's symptoms, pulmonary function, and daytime and nocturnal gas exchange. The aim is to begin when there are symptoms with significant pulmonary function and/or gas-exchange abnormalities, which is when the patient still has time to adapt but well before the occurrence of a respiratory crisis.

## Central Hypoventilation/ Obstructive Sleep Apnea

The first case reports describing the use of nasal ventilation for chronic respiratory failure were in young children with central hypoventilation, who had resolution of gas-exchange abnormalities and symptoms after initiation of therapy.[248,249] No controlled studies have examined this application, but enough anecdotal evidence has accrued that consensus groups consider therapy of central hypoventilation as an appropriate indication for NIPPV.[200,238]

Nasal CPAP is considered the therapy of first choice for obstructive sleep apnea. NIPPV, however, may be successful in improving daytime gas exchange and symptoms in hypoventilating patients with obstructive sleep apnea who have persistent $CO_2$ retention after use of nasal CPAP alone. Among thirteen patients with severe obstructive sleep apnea whose hypercapnia (average $Pa_{CO_2}$ of 62 mm Hg) was unresponsive to CPAP, NIPPV using volume-limited ventilators lowered the $Pa_{CO_2}$ to 46 mm Hg, and nine of the patients eventually stabilized on CPAP alone.[250]

Independently adjusted inspiratory and expiratory pressure or "bilevel" positive-pressure ventilation was first developed as a way of controlling obstructive sleep apnea while using lower expiratory pressures than with CPAP alone, thus potentially enhancing comfort and adherence with therapy.[251] Reeves-Hoche et al[252] were unable to demonstrate improved adherence rates in patients with obstructive sleep apnea treated with bilevel ventilation compared with CPAP alone. Even so, bilevel devices are still used commonly to treat patients intolerant of CPAP alone, but a stronger rationale supports the use of bilevel NIPPV for obstructive apnea if patients have persisting hypoventilation despite adequate CPAP therapy.

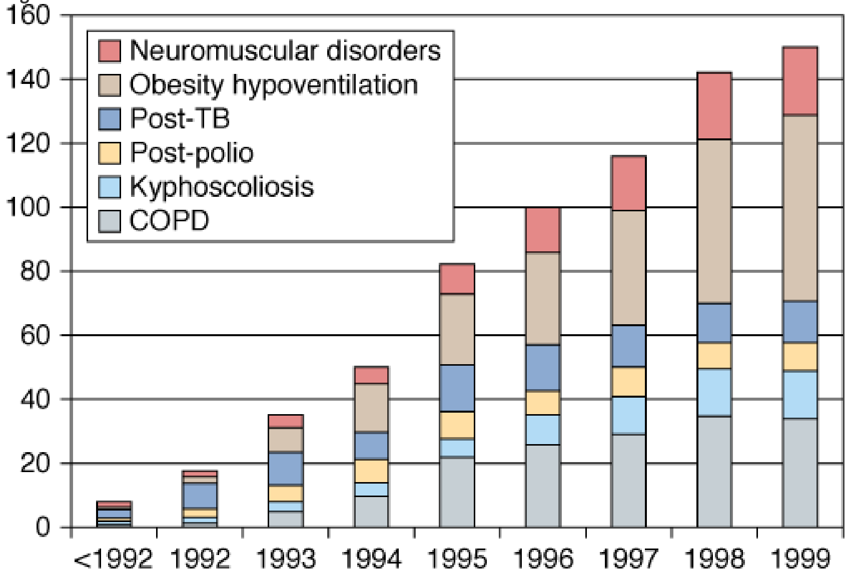
## Obesity-Hypoventilation Syndrome

Respiratory impairment is common among obese patients, including those with restricted lung volumes secondary to increased chest wall and lung elastance, abnormal blood gases, and breathing disturbances during sleep. When obese patients hypoventilate, the term *obesity-hypoventilation syndrome* is applied, a condition that is multifactorial in etiology. The altered chest wall mechanics are accompanied by reductions in respiratory drive (either acquired or congenital), as well as obstructive sleep apnea (in 80% to 90% of patients), giving rise to the hypoventilation.[253] This is a morbid condition associated with cor pulmonale and a high mortality rate over time, but it responds favorably to NIV.

Morbidly obese subjects have significantly increased work of breathing at baseline, and NIPPV (inspiratory pressure of 9 to 12 cm $H_2O$, expiratory pressure of 4 cm $H_2O$) lowers that work.[254] NIPPV also raises tidal volumes and lowers end-tidal $P_{CO_2}$ more in patients with obesity hypoventilation than in obese patients without sleep-disordered breathing or in nonhypoventilating subjects with obstructive sleep apnea.[254] NIPPV lowers and improves symptoms as effectively in patients with obesity hypoventilation as in those with severe kyphoscoliosis,[255] associated with an increase in respiratory drive.[255] CPAP alone, however, may be as effective as NIPPV in some patients.[253] Some investigators also have started with NIPPV and converted to CPAP once hypoventilation has been controlled.[250] In a recent randomized controlled trial of thirty-seven patients with obesity hypoventilation and mild hypercapnia, 1 month of NIV (inspiratory and expiratory pressures 18 and 11 cm $H_2O$, respectively; backup rate: 13 breaths/min), eighteen NIV patients had improved sleep architecture and gas exchange, but indices of inflammation, endothelial function, and arterial stiffness did not improve compared to a control group given lifestyle advice.[256] These studies support the use of NIPPV for obesity hypoventilation to improve symptoms, sleep quality, and gas exchange.

As obesity has become an increasingly prevalent problem, obesity-hypoventilation syndrome has become an increasingly common indication for using long-term NIPPV. The Swiss survey found that during the 1990s it became the most common diagnosis among patients using long-term NIPPV at home in the Geneva region (Fig. 18-7).[230] Adherence was excellent with the therapy, exceeding 70%, average $Pa_{CO_2}$ normalized, and hospital days fell significantly following NIPPV initiation. A subsequent large cohort of obesity-hypoventilation patients from France was followed long term and manifested favorable sustained responses to NIPPV, with a 3-year continuation rate of 80%, sustained improvements in gas exchange, and a 5-year survival of 77.3%.[257]

Figure 18-7

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Yearly count of the cumulative population of patients treated by noninvasive positive-pressure ventilation during the study period (1992 to 2000) according to diagnostic category. (Reproduced with permission from the American College of Chest Physicians from Janseens, et al. Changing patterns in long-term noninvasive ventilation: a 7-year prospective study in the Geneva Lake area. *Chest.* 2003;123:67–79.)

## Selection of Patients with Central Sleep Apnea/Obesity-Hypoventilation Syndrome for Noninvasive Positive-Pressure Ventilation

Firm indications and selection guidelines for central hypoventilation/central sleep apnea/obesity-hypoventilation syndrome have not been established although the Canadian Thoracic Society consensus is that NIPPV is the treatment of choice unless $CO_2$ retention is no more than mild, in which case CPAP alone might suffice.[239] Patients who are symptomatic with frank hypoventilation or who have prolonged central apneas clearly are good candidates for NIPPV. The more severe the hypoventilation, the more important it would be to start with a mode that augments minute volume, and using a backup rate would be important for those with prolonged central apneas. Medicare guidelines in the United States (Table 18-6) require a polysomnogram to document the central apneas, prolonged $O_2$ desaturations (≤88% for more than 5 minutes) as evidence of nocturnal hypoventilation and, if the patient has obstructive sleep apnea, evidence of CPAP failure with improvement on NIPPV, as determined by oximetry or a repeat polysomnogram.

**Table 18-6: Guidelines for Noninvasive Positive-Pressure Ventilation in Obstructive Sleep Apnea/Central Sleep Apnea***

1. Polysomnogram demonstrating OSA, CSA, or mixed apneas
2. If OSA, patient failed to improve or tolerate CPAP alone
3. Sustained oxygen desaturation nocturnally (≤88% for >5 min)
4. Significant improvement in nocturnal gas exchange during NIPPV use, as documented by oximetry or polysomnography

*Abbreviations: CSA,* central sleep apnea; *NIPPV,* noninvasive positive-pressure ventilation; *OSA,* obstructive sleep apnea.

*Based on Center for Medicare and Medicaid Services Guidelines.

# Obstructive Lung Diseases

## Chronic Obstructive Pulmonary Disease

During the 1960s, McClement et al[258] speculated that improved ventilation–perfusion relationships, reduced $O_2$ consumption, and mobilization of secretions would benefit patients with COPD who were treated with negative-pressure ventilators. During the early 1980s, Braun and Marino[259] tested the theory that respiratory muscles are chronically fatigued and will benefit from intermittent rest. They treated sixteen patients with severe COPD using wrap negative-pressure ventilators for 5 hours daily over 5 months and observed improvements in vital capacity, maximal inspiratory and expiratory pressure, and daytime $Pa_{CO_2}$ during spontaneous breathing. Although these results were interpreted as supporting the "muscle rest" hypothesis, controls were lacking, and other aspects of rehabilitation or passage of time alone could have been responsible for the improvements.

Subsequently, several controlled studies showed improvement in respiratory muscle strength after short-term (days to a week) use of negative-pressure ventilation.[260–262] These studies documented respiratory muscle rest by showing significant reductions in the diaphragmatic electromyographic signal. Subsequent controlled trials, however, of longer duration (up to several months) showed no benefit.[263–265] Notably, baseline $Pa_{CO_2}$ among the latter unfavorable trials was approximately 47 mm Hg, substantially lower than that in the favorable studies (57 mm Hg). This raises the possibility that respiratory muscles in patients with severe $CO_2$ retention are more likely to benefit from intermittent negative-pressure ventilation than patients with little or no $CO_2$ retention, perhaps because of relief of the unfavorable effect of hypercapnia on respiratory muscle function.[266]

Negative-pressure ventilation was tolerated poorly in the preceding trials, so subsequent trials tested NIPPV to see if better tolerance might achieve more consistent benefits. In addition, patients with severe COPD are known to have more frequent nocturnal desaturations related to hypoventilation than normal subjects. These desaturations are associated with arousals that shorten the duration and diminish the quality of sleep, an effect that can be ameliorated by $O_2$ supplementation, at least in "blue and bloated" patients.[267] Furthermore, patients with COPD have a 32% drop in inspiratory flow rate during rapid eye movement sleep that is associated with a reduced tidal volume.[268] By assisting ventilation, NIPPV offers the potential of restoring inspiratory flow, eliminating episodes of hypoventilation, and improving nocturnal gas exchange, as well as the duration and quality of sleep.

Studies using NIPPV in patients with severe obstructive lung diseases have yielded conflicting results. Initial small uncontrolled cohort series on the use of nasal NIPPV in patients with severe stable COPD lent support to the idea that NIPPV would improve sleep efficiency and daytime and nocturnal gas exchange.[269,270] A 3-month crossover trial by Strumpf et al,[36] however, found improvement only in neuropsychological function but not in nocturnal or daytime gas exchange, sleep quality, pulmonary functions, exercise tolerance, or symptoms. This study also encountered a high dropout rate, with seven patients withdrawing because of mask intolerance and only seven of nineteen entered patients actually completing the trial. In contrast, in a study of nearly identical design, Meecham-Jones et al[271] enrolled eighteen patients with severe COPD, fourteen of whom completed the study. Nocturnal and daytime gas exchange, total sleep time, and symptoms improved during NIPPV use. These salutary effects of NIPPV on sleep duration and efficiency in patients with severe stable COPD also were observed in a 2-night crossover trial in six patients with an initial $Pa_{CO_2}$ of 58 mm Hg.[272] Some of the disparity between these studies may be explained by the observation that patients in the study of Strumpf et al had more severe airway obstruction ($FEV_1$ of 0.54 vs. 0.81 L) despite having less $CO_2$ retention ($Pa_{CO_2}$ of 47 vs. 57 mm Hg) than did patients in the study of Meecham-Jones et al. This suggests that different subsets of patients with COPD were entered into the studies and that those with greater $CO_2$ retention ("blue bloaters") may be more likely to benefit from NIPPV.

Two other randomized, controlled trials failed to substantiate the hypothesis that greater $CO_2$ retention predicts NIPPV success in patients with severe COPD, despite attempts to enroll hypercapnic subjects. Gay et al[273] screened thirty-two hypercapnic patients, but only thirteen remained after exclusion for obstructive sleep apnea or other terminal illness. Only four of the seven patients randomized to NIPPV completed the trial and, not surprisingly, no significant differences emerged. Lin et al[274] performed an 8-week crossover trial consisting of consecutive, randomized 2-week periods of no therapy, $O_2$ alone, NIPPV alone, and NIPPV combined with $O_2$. Among twelve patients with a mean initial $Pa_{CO_2}$ of 50.5 mm Hg, NIPPV not only conferred no added benefit over $O_2$ alone with regard to oxygenation, ventricular function, or sleep quality, but it also reduced sleep efficiency and total sleep time. The authors, however, used inspiratory pressures of only 12 cm $H_2O$, which may have provided insufficient ventilator assistance, and 2 weeks may have been too brief to permit adequate adaptation to NIPPV.

Several longer-term controlled trials have been performed subsequently. Casanova et al[275] performed a randomized year-long trial in forty-four patients with severe COPD, finding no improvements in gas exchange, survival, or hospitalization rate, although one test of neuropsychological function improved. This study, however, also used relatively low inspiratory pressures and did not assess sleep quality or health status. In an Italian multicenter trial, Clini et al[276] screened 120 patients with severe COPD and chronic $CO_2$ retention ($Pa_{CO_2}$ of 50 mm Hg or more). Ninety patients were enrolled. After dropouts and deaths, forty-seven were left, divided between the NIPPV plus $O_2$ and $O_2$ alone groups. Patients treated with NIPPV had less of an increase in $Pa_{CO_2}$ over the 2-year period than controls (55 to 56 vs. 55 to 60 mm Hg, respectively, $p < 0.05$), less deterioration in the Maugeri Respiratory Failure (MRF-28) functional score (although the St. George's Respiratory Questionnaire was no different), and a trend toward fewer hospital days per patient per year (20 days before and 14 days after initiation of NIPPV). No differences were detected in the 6-minute walk distance, dyspnea score, sleep symptoms, or mortality rate. An Australian trial of 144 patients with severe COPD and baseline $Pa_{CO_2}$ is in the low 50 to 53 mm Hg randomized patients to NIPPV or long-term oxygen therapy.[277] The NIPPV group had improved adjusted (OR: 0.63; confidence interval [CI]: 0.40 to 0.99) but not unadjusted survival, and a worse score on the St. George's health status scale for general and mental health. There were no differences in gas exchange after the first 6 months. Only forty-one of the seventy-two patients randomized to NIPPV used it for more than 4 hours nightly and were included in the survival analysis, and average inspiratory and expiratory pressures were 12.9 and 5.1 cm $H_2O$, respectively.

These studies have lacked statistical power and results have been inconsistent, with some showing benefit only for physiologic variables such as respiratory muscle strength and $Pa_{CO_2}$ or total sleep time. Some have shown improved mortality, functional status and quality of life, but most have not. Furthermore, continuation rates have been relatively low compared to neuromuscular or chest wall disorders (see Fig. 18-7). This suggests that patients with COPD are less tolerant of or benefit less from NIPPV than neuromuscular or obesity hypoventilation patients. Criner at al[278] initiated NIPPV in twenty patients with neuromuscular disease and twenty with COPD during a several-week stay in a specialized ventilator unit. Despite these optimal conditions, only 50% of patients with COPD as compared with 80% of those with neuromuscular disease were still using NIPPV after 6 months.

One concern has been that ventilator pressures have been too low in many of the studies to adequately enhance gas exchange. Windisch et al[279] have pioneered the use of "high-intensity" NIPPV to allay these concerns in patients with COPD. In a cohort of seventy-three patients, average inspiratory pressure of 28 mm Hg was used, and patients had a relatively low hospitalization rate (22%) during the subsequent year and higher-than-expected survival at 3 and 5 years (82% and 58%, respectively). In a follow-up study, Dreher et al[280] performed a 6-week crossover trial comparing high-intensity versus low-intensity NIPPV (average inspiratory pressure: 28.6 and 14.6 cm $H_2O$, respectively). At the end of the trial, the high-intensity group used NIPPV 3.6 hours more per day than did the low-intensity group, and there were significant improvements in exercise-related dyspnea, daytime $Pa_{CO_2}$, $FEV_1$, vital capacity, and the Severe Respiratory Insufficiency Questionnaire Summary score.

Overall, the results of these long-term trials testing the efficacy of NIPPV in severe stable COPD have been disappointing, and this application remains controversial.[281] A meta-analysis of the earlier controlled trials concluded that the studies were too small to discern a "clear clinical direction."[282] It should be acknowledged that three subsequent randomized trials[276,277,280] yielded favorable findings, although all have methodologic limitations. Despite controversy about the strength of the evidence for benefit, NIV is used widely in chronic stable COPD in some countries.[283] The high-intensity approach advocated by Windisch et al also shows promise, but requires further validation at other centers.

## Noninvasive Positive-Pressure Ventilation to Enhance Rehabilitation in Chronic Obstructive Pulmonary Disease

NIPPV may serve as an adjunct to exercise training in rehabilitation for patients with severe stable COPD. Two different approaches have been used: one employs NIPPV to unload the inspiratory muscles during exercise and to permit a greater exercise intensity to magnify the training effect; the other rests muscles between sessions (mainly at night) to enhance daytime function during the sessions. Investigations show that CPAP and PSV singly and in combination increase exercise capacity in patients with severe COPD.[284,235] Bianchi et al[286] showed that compared with CPAP or PSV, proportional-assist ventilation brought about the greatest improvement in cycling endurance and reduction in dyspnea in fifteen stable hypercapnic patients with COPD. This enhanced exercise capacity during ventilator use, however, has not yet been shown to translate into a greater training effect or functional improvement during spontaneous breathing.[287] Garrod et al[288] tested the second approach among forty-five patients with severe COPD ($FEV_1$ < 50% of predicted), showing that nocturnal NIPPV between rehabilitation sessions increased the shuttle-walk distance and improved quality of life compared with standard therapy. Duiverman et al[289] tested a similar approach, randomizing seventy-two patients with severe COPD to nocturnal NIPPV plus rehabilitation and rehabilitation alone. The combination group had greater improvements in some domains of quality of life as well as daytime $Pa_{CO_2}$, daily step count, and minute ventilation. These studies indicate that NIPPV, used either during or between exercise sessions, has the potential to enhance benefits accruing from pulmonary rehabilitation, but more confirmatory studies are needed.

## Cystic Fibrosis and Diffuse Bronchiectasis

Small case series[290,291] have reported stabilization and sometimes even improvement of gas-exchange abnormalities for periods ranging up to 15 months in severely hypercapnic ($Pa_{CO_2}$ > 54 mm Hg) patients with end-stage cystic fibrosis awaiting lung transplantation. Gozal et al[292] observed markedly improved gas exchange in all sleep stages in six patients with cystic fibrosis treated with NIPPV plus $O_2$ therapy in comparison with patients treated with $O_2$ therapy alone, although sleep duration and architecture were similar in the two conditions. Cystic fibrosis is now a common reason among children for the use of NIPPV at home, constituting 17% of such children in a French survey.[293]

The mechanism by which NIPPV assists cystic fibrosis patients is not entirely clear. NIPPV reduces the work of breathing significantly,[294] but a study in thirteen hypercapnic patients with cystic fibrosis (average $Pa_{CO_2}$ of 51 mm Hg) found no improvements in sleep quality, daytime arterial blood gases, pulmonary function tests, respiratory muscle strength, or exercise tolerance after 2 months of NIPPV, even though eight of the patients felt improved symptomatically.[295] Madden et al[296] found improved hypoxemia but again no improvement in hypercapnia among 113 patients treated long term with NIPPV; these authors still considered NIPPV useful as a bridge to lung transplantation. NIPPV also can be useful for administration of aerosol to cystic fibrosis patients. Fauroux et al[297] found that it was superior to a standard nebulizer in a small group of cystic fibrosis patients. Thus, NIPPV appears to have a role in supporting deteriorating patients with cystic fibrosis and serving as a bridge to transplantation, even though improvements in gas exchange and sleep parameters are not seen consistently.

In diffuse bronchiectasis patients, Benhamou et al[298] found that the use of NIPPV was associated with improved Karnovsky function scores and a reduction in days of hospitalization from 46 days for the year before to 21 days for the year after starting NIPPV. Compared with a historical control group, however, rates of deterioration in oxygenation were similar, and no survival benefit was apparent. In fact, in the long-term English follow-up study,[231] patients with end-stage bronchiectasis had poorer survivals than other patient subgroups, most dying within 2 years. Dupont et al[299] retrospectively reviewed the outcomes of forty-eight patients with diffuse bronchiectasis following their first ICU admission over a 10-year period; 27% were treated with NIPPV and 54% required intubation. One-year mortality was 40%. Age older than 65 years, a higher simplified acute physiology score II score

(>32), and the need for intubation were identified as predictors of mortality. These studies suggest a role for NIPPV in treating patients with cystic fibrosis and diffuse bronchiectasis who have developed severe $CO_2$ retention, as well as in serving as a bridge to transplantation, but the capacity to prolong life may be limited.

A recent Cochrane analysis[300] concluded that NIPPV may be helpful in facilitating secretion clearance in patients with cystic fibrosis and when combined with $O_2$ supplementation, may improve nocturnal gas exchange, but cited the lack of controlled trials as a limitation in making recommendations. Lacking such controlled trials, definitive recommendations on how to select patients with cystic fibrosis or diffuse bronchiectasis for NIPPV or when to start are unavailable; most clinicians use guidelines similar to those used for severe COPD (see below), paying particular attention to the inclusion of techniques to aid in secretion clearance.

## Selection of Patients with Chronic Respiratory Failure and Obstructive Lung Diseases to Receive Noninvasive Ventilation

An earlier consensus statement noted the discordant results of the available trials and concluded that more study is needed before NIPPV can be recommended in severe stable COPD.[200] A subsequent consensus conference agreed that the data are scanty and conflicting but opined that the available evidence suggests that certain subgroups of patients with COPD may benefit.[238] Most trials that have observed benefit from either negative-pressure or positive-pressure NIV in severe stable COPD have enrolled patients with more $CO_2$ retention at baseline than trials with negative results. Thus, the consensus opinion was that a trial of NIPPV in severe stable COPD patients is justified if $CO_2$ retention is severe (i.e., $Pa_{CO_2}$ > 55 mm Hg).

Considering that one of the four controlled trials reporting beneficial effects of NIPPV in severe stable COPD patients[271] enrolled patients with frequent hypopneas (ten per hour) and $O_2$ desaturations during sleep, another indication suggested by the consensus group was sustained, severe nocturnal $O_2$ desaturation (< 88% for more than 5 consecutive minutes). $O_2$ therapy alone, however, has been shown to improve sleep quality, reduce drowsiness, and improve neuropsychological function in such patients.[267,268] Therefore, the recommendation was made that sleep monitoring be performed during $O_2$ supplementation and that NIPPV not be initiated unless symptoms fail to respond to a trial of long-term $O_2$ therapy. If patients have a daytime $Pa_{CO_2}$ between 50 and 54 mm Hg, the consensus group opined that NIPPV should be used if such patients have evidence of nocturnal hypoventilation, as indicated by nocturnal oximetry, or if there is a history of repeated hospitalizations. Patients whose $CO_2$ retention worsens substantially during $O_2$ therapy should also be considered for NIPPV, because such patients responded favorably to NIPPV in an uncontrolled trial.[301]

In the absence of more controlled trials with favorable findings, however, these guidelines are tentative. Also, even for patients who meet the criteria, patient tolerance of NIPPV may be poor.[278] To maximize patient compliance, only motivated symptomatic patients, such as those with fatigue or daytime hypersomnolence, who can cooperate and comprehend the purpose of the therapy should be selected. As outlined in Table 18-7, NIV should not be initiated unless other therapies have been optimized, including $O_2$ supplementation and CPAP (if indicated). These guidelines have led to reduced use of NIPPV for severe stable COPD in the United States since the late 1990s, when certain home respiratory companies were encouraging widespread use. Use is currently more prevalent in certain European countries, such as Switzerland, where a survey found that COPD was the second most common reason for use of NIPPV in the home.[230] A pan-European survey on home mechanical ventilation showed enormous variability between countries in the proportion of patients receiving ventilation for neuromuscular versus lung diseases and between those receiving noninvasive versus tracheostomy ventilation.[283]

**Table 18-7: Recommended Guidelines for Selection of Patients with Obstructive Lung Diseases to Receive Long-Term Noninvasive Positive-Pressure Ventilation***

1. Symptoms: fatigue, hypersomnolence, dyspnea, and so on
2. Failure to respond to optimal medical therapy: Maximal bronchodilator therapy and/or steroids $O_2$ supplementation if indicated
3. Gas-exchange abnormalities:
   $Pa_{CO_2} \geq 52$ mm Hg
   $Sa_{O_2} < 88\%$ for more than 5 consecutive minutes nocturnally despite $O_2$ supplementation
4. Obstructive sleep apnea excluded on clinical grounds or failure to respond to CPAP therapy if moderate to severe obstructive sleep apnea detected on sleep study
5. Reassess after 2 months' therapy; continue if adequate compliance (>4 hours a day) and favorable therapeutic response

*Based on Center for Medicaid and Medicare Services reimbursement guidelines.

## Congestive Heart Failure

As discussed earlier, evidence supports the use of NIV (either CPAP alone or NIPPV) in the therapy of acute heart failure, and it also may have a role in chronic CHF. Increases in intrathoracic pressure have long been known to have salutary hemodynamic effects in some patients with congestive heart failure. Naughton et al[21] found that CPAP (10 cm $H_2O$) reduced both ventilatory work (by minimizing negative intrathoracic pressure swings) and cardiac load (by reducing transmural pressure) in fifteen patients with congestive heart failure. A subsequent study found that longer-term nocturnal CPAP (9 cm $H_2O$) improved inspiratory muscle strength (maximal inspiratory pressure increased from 79.3 to 90.7 cm $H_2O$) in a group of eight patients with CHF.[302] One month of nocturnal CPAP also increased left-ventricular ejection fraction (33.8% vs. 25%) and lowered systolic systemic pressure (116 vs. 126 mm Hg) in patients with CHF and obstructive sleep apnea compared with healthy subjects.[303]

Whether NIPPV is better than CPAP alone in these patients has been controversial. Willson et al[304] found dramatic improvements in sleep parameters (apnea-hypopnea index: 49–6; arousal index: 42–17) in patients with CHF and Cheyne-Stokes respiration after treatment with a portable bilevel device. Conversely, Kohnlein et al[305] performed a crossover trial consisting of randomized 2-week periods of NIPPV and CPAP in thirty-five patients with Cheyne-Stokes respiration. Both modalities improved apnea-hypopnea and arousal indexes dramatically, but there was no difference between the two. Teschler et al[306] randomized fourteen patients with CHF and Cheyne-Stokes respiration to control, $O_2$ alone, CPAP alone, bilevel NIPPV, and adaptive pressure-support servo ventilation on five separate randomly ordered nights. They observed equal reductions in apnea-hypopnea and arousal indexes with $O_2$ alone and CPAP alone, a greater reduction with bilevel ventilation, and the greatest improvement with adaptive pressure-support servo ventilation. This study supports the idea that customized modes designed to respond to the apneas of Cheyne-Stokes respiration (such as adaptive pressure-support servo ventilation) may be especially effective. But no adequately powered trials addressing important clinical outcomes have yet been performed.

The Canadian CPAP (CanPAP) trial randomized 258 patients with CHF and central sleep apnea to receive CPAP or no CPAP. CPAP attenuated central sleep apnea, improved nocturnal oxygenation, increased the left ventricular ejection fraction, lowered norepinephrine levels, and increased the distance walked in six minutes by 21 meters. It did not, however, improve quality of life or survival and the authors concluded that their data "do not support the use of CPAP to extend life" in patients with CHF and central sleep apnea.[307] In post hoc analysis, however, these authors subsequently reported that in a subgroup of patients whose apnea-hypopnea index was lowered to below 15/hour by CPAP, transplantation-free survival was significantly improved.[308] Thus, the role of NIPPV in patients with CHF is currently unclear. Most clinicians currently use CPAP alone for patients with CHF and obstructive sleep apnea and optimal medical therapy, including $O_2$ supplementation, for those with Cheyne-Stokes breathing. Some favor adaptive pressure-support servo ventilation as the preferred noninvasive mode for patients with CHF and "complex" sleep apnea characterized by periodic central apneas (Cheyne-Stokes breathing),[309] but further studies are needed before firm recommendations can be made.

## Pediatric Uses of Noninvasive Positive-Pressure Ventilation for Chronic Respiratory Failure

Since the first case reports on the successful use of nasal NIPPV in children with central hypoventilation,[199,200] relatively few reports of NIPPV have appeared in the pediatric literature. Nonetheless, some of the experience in adults can be applied to children because such conditions as Duchenne muscular dystrophy or cystic fibrosis may impair respiratory function in older children, and these have been included in a number of the published reports.[248,249] In their experience with fifteen children having neuromuscular disease or cystic fibrosis treated with nasal NIPPV and followed for periods ranging from 1 to 21 months, Padman et al[184] found that average $Pa_{CO_2}$ and hospital utilization fell; only one child required an artificial airway. Fauroux et al[310] undertook a French survey on the use of NIPPV by children at home. Of 102 children followed at fifteen centers, 7% were younger than 3 years of age, 35% were 4 to 11 years of age, and 58% were 12 years of age, and 34% had neuromuscular disease, 30% had obstructive sleep apnea or craniofacial abnormalities, 17% had cystic fibrosis, 9% had central hypoventilation, and 8% had scoliosis. In a subsequent report, Fauroux et al[311] described flattening of the face in 48% of patients. Nevertheless, pediatric patients appear to respond as well to NIPPV as most adults with chronic respiratory failure. In a long-term follow-up study of thirty pediatric patients (average age: 12.3 years) with mainly non-Duchenne neuromuscular syndromes, Mellies et al[312] observed clinical stability exceeding an average of 2 years in duration. Nocturnal and diurnal gas exchange, quality of sleep, and symptoms were improved, and these deteriorated promptly on temporary withdrawal of NIPPV. The authors concluded that NIPPV is effective and should be used in children with symptomatic sleep-disordered breathing associated with neuromuscular syndromes.

## Practical Application of Noninvasive Positive-Pressure Ventilation

Despite the accumulating evidence on NIPPV indications that helps in selecting appropriate patients, the delivery of NIPPV remains very much an art. After the decision is made to treat a patient with NIPPV, the clinician must decide on a mask (or interface), ventilator, settings, and adjuncts. NIPPV must be delivered in a safe and adequately monitored location. Implementation of each step requires knowledge and experience. More than with invasive ventilation, the interaction between patient and clinician is central to success. The following will provide an overview of the steps in this process.

### Initiation

Although little scientific evidence is available to guide the decisions surrounding initiation, they should be made carefully because success or failure depends on them. Focusing on the major goals of NIV may help (Table 18-8). NIV shares with invasive ventilation the goals of improving gas exchange, either nocturnal, daytime, or both, and minimizing complications. Even more than with invasive ventilation, though, NIV seeks to alleviate symptoms and optimize comfort. Because of the open-circuit design of noninvasive positive-pressure ventilators, success depends largely on patient cooperation and acceptance. Patient tolerance is an important goal because the other goals are not achievable unless the patient accepts the therapy. The goals of acute and long-term applications are overlapping, but alleviation of increased work of breathing is an important goal in acute applications, whereas improvement in sleep duration and quality is more important during long-term applications. The following gives recommendations for initiation, citing evidence when available, pointing out controversy where it exists, and offering opinion where necessary. Most sections offer comments on both acute and long-term applications.

Table 18-8: Goals of Noninvasive Ventilation

| |
|---|
| *Acute applications* |
| Relieve dyspnea |
| Optimize patient comfort |
| Reduce work of breathing |
| Improve or stabilize gas exchange |
| Minimize complications |
| Avoid intubation |
| Avoid delay of needed intubation |
| *Long-term applications* |
| Ameliorate symptoms |
| Improve or stabilize gas exchange |
| Improve sleep duration and quality |
| Maximize quality of life |
| Enhance functional status |
| Prolong survival |

*Source:* Reprinted with permission of the American Thoracic Society. Copyright © 2012 American Thoracic Society. From Mehta S, Hill NS. Noninvasive ventilation: state of the art. *Am J Respir Crit Care Med.* 2001; 163:540–577. Official Journal of the American Thoracic Society.

## Location

### Acute

NIV can be initiated wherever the patient presents with acute respiratory distress—in the emergency department,[60,313] ICU,[17,61] intermediate-care or respiratory-care unit, or hospital ward.[62,314] A survey of acute care hospitals in Massachusetts and Rhode Island found that a third of NIPPV initiations were in the emergency department, and half were in the ICU.[315] Following initiation, transfer to a location that offers continuous monitoring is recommended until the patient stabilizes. The patient's acuity of illness and risk of deterioration if an accidental disconnection occurs should dictate the intensity of monitoring. One study used 15 minutes of stability following discontinuation of NIPPV as a criterion for admission to a regular ward,[316] but a longer period of time (30 minutes for example) may be safer. During transfers, ventilator assistance and monitoring should be continued because rapid deteriorations can occur.

NIPPV is used on regular wards in many hospitals because of the scarcity of ICU beds, but some guidelines have recommended that NIPPV be applied only in the ICU because of concerns about patient safety.[317] Others have suggested that if pH < 7.30 in patients with COPD, ICU transfer is advisable. Farha et al[318] reported on their experience with seventy-six patients treated on a regular ward with NIPPV. Of the sixty-two patients without a do-not-intubate status, 31% required intubation and were transferred to the ICU. The authors considered this comparable to the experience with patients treated in more closely monitored settings, concluding that NIPPV can be administered safely on regular floors. But unless the ward has considerable experience administering NIPPV, only stable patients should be treated there.

### Chronic

Stable patients with chronic respiratory failure may start NIPPV during an inpatient admission, in a sleep laboratory during the daytime or an overnight stay, in a physician's outpatient office (with therapists from the home respiratory vendor present), or at home. In a randomized controlled trial of twenty-eight patients, mainly with neuromuscular disease, patients initiated at home experienced just as good outcomes after 2 months as those begun as inpatients, including gas exchange and ventilator adherence, except that an average 3.8-day hospitalization was avoided.[319] Thus, routine hospitalization is unnecessary unless warranted by the patient's medical condition. Initial use of a sleep laboratory offers the advantage of precise titration of initial pressure or volume settings during sleep monitoring but adds to costs and may delay implementation because of scheduling problems. Also, no titration protocol has been validated, and selecting pressures to eliminate apneas and hypopneas, as is done with sleep apnea, may not be adequate to reverse hypoventilation in patients with chronic respiratory failure. Until outcome studies demonstrate the superiority of one location over another, the choice of location will be based on practitioner preference. Perhaps more important than the specific location is the availability of skilled, attentive practitioners to help during the initiation and adaptation processes.

## Masks (Interfaces)

A daunting array of interfaces has become available to deliver NIPPV, and detailed descriptions can be found elsewhere.[320] In brief, the most commonly used interfaces in both acute and long-term settings are nasal and oronasal (or full-face) masks. Nasal masks usually are triangular clear plastic domes that have soft silicone sealing surfaces. Oronasal (or full-face) masks are similar in appearance but are larger and fit over the nose and mouth. Nasal interfaces offer many modifications, however, including nasal pillows with soft rubber cones that insert directly into the nares, so-called minimasks that fit over the tip of the nose, and gel-filled seals designed to enhance comfort. Various oronasal masks are available, including those with foam-filled or air-filled seals and a chin support. A larger version of the full-face mask is available that seals around the perimeter of the face, potentially enhancing comfort and eliminating the development of nasal bridge ulcers.[321] Oral interfaces also are used occasionally, mainly in the long-term setting in patients with neuromuscular disease.[205]

More recently, a number of studies have evaluated the "helmet," a novel interface for NIV that consists of a clear plastic cylinder that fits over the head and seals on the shoulders. The Food and Drug Administration has not yet approved this application in the United States. It avoids contact with the nose and mouth, eliminating nasal ulceration and potentially enhancing comfort.[322] CPAP delivered via the helmet to patients with acute respiratory failure is better tolerated than the full-face mask in historically matched controls.[323] Compared with the full-face mask used to deliver PSV in patients with COPD in acute respiratory failure[324] the helmet similarly improved vital signs, achieved similar intubation and mortality rates, and reduced complications. A more recent randomized trial showed that the Helmet is less efficient than a full face mask at $CO_2$ removal and can cause problems with triggering and cycling during pressure-support ventilation.[325] $Pa_{CO_2}$ tends to be higher in patients treated with the helmet, raising concerns about rebreathing. High flow rates must be used to avoid this problem,[326] but this contributes to noise levels that may exceed 100 dB.[327]

Oral interfaces have been used successfully for many years in patients with slowly progressive neuromuscular diseases.[205] Long-term nasal ventilation appears to offer improved tolerance compared with mouthpiece ventilation in some patients.[211] Either may be effective, even in patients with minimal pulmonary reserve, and both may be used in the same patient, nasal ventilation during sleep at night, for example, with mouthpiece ventilation used as needed during the daytime.[211]

## Selection of Interfaces

### Acute

Ideally, interfaces for the acute setting should be inexpensive and disposable or reusable without sacrificing comfort. A randomized, controlled trial in seventy patients with acute respiratory failure showed that full-face and nasal masks similarly improve dyspnea, vital signs, and gas exchange, but the nasal mask had a higher initial intolerance rate (34% vs. 12%), attributed to air leaking through the mouth.[328] Mouth leaks have been reported to occur in as many as 94% of patients receiving NIPPV for hypercapnic acute respiratory failure and commonly contribute to mask failures.[329] Thus, the full-face mask is usually the first choice in the acute setting, although claustrophobic patients or those with a need to expectorate frequently may fare better with nasal masks. In a recent randomized trial, the larger Total-face mask that seals around the perimeter of the face was equivalent to the standard full-face mask with regard to patient comfort and NIPPV failure rate, but some patients declined early on to use it because of its imposing appearance.[330] Concerns have been raised about the dead space attributable to the large volumes of these face masks, but "streaming" of airflow directly from the inlet to the patient's nose and mouth appears to minimize this problem.[331,332]

The oral interface is used sometimes in the acute setting to facilitate initial patient adaptation.[333] In a comparison study, however, of four different NIPPV interfaces in patients with acute respiratory failure, the mouthpiece had the largest leak, while there was no significant difference between the Total-face mask, oronasal mask, or nasal mask.[334] In a short-term randomized crossover evaluation in healthy volunteers, higher pressures (15 inspiratory, 10 expiratory cm $H_2O$) were more uncomfortable than lower pressures (10 inspiratory, 6 expiratory cm $H_2O$) regardless of mask; the Total-face mask avoided nasal bridge discomfort and has less rebreathing, the nasal mask caused less oral dryness, and the standard full-face mask was associated with bothersome air leaking into the eyes.[335]

The above findings emphasize that many different mask types are available, all of which may be acceptable depending on caregiver preferences, proper fitting, ventilator settings, individual patient characteristics, and other factors. Thus, when initiating NIPPV, clinicians should have a variety of interfaces readily available; a "mask bag" can be suspended from the ventilator so that individual patient needs can be accommodated.

### Chronic

Comfort and tolerance are even more important in the long-term setting because interfaces must be used for months, mainly during sleep, before being replaced. Many different interfaces are available partly because of the demand driven by the large population of patients with obstructive sleep apnea using similar technology. Standard nasal masks are the most commonly used interfaces in the chronic setting, and a short-term controlled trial on naive patients with restrictive and obstructive forms of chronic respiratory failure found that patients rated these nasal masks as more comfortable than nasal prongs or full-face masks.[336] A polysomnographic comparison of the two masks in patients with chronic respiratory failure[337] found that nasal and full-face masks were equivalent with regard to apnea-hypopnea index, gas exchange, and sleep quality, but the nasal mask required chin straps to control mouth leaks, and sleep efficiency was less with the full-face mask.

Once again, clinicians must be prepared to try a number of different interfaces to optimize comfort. Fitting gauges should be used when available to facilitate proper sizing, and strap tension should be the minimum that controls leaks. Headstrap materials, tightness, and attachments to the head and mask are also important for comfort. Many different types of headstraps are available, although they usually are designed for a particular mask.

## Selection of a Ventilator

### Acute

A steadily expanding number of ventilators is available for NIPPV in the acute setting. Bilevel devices are portable PSV ventilators, first developed for home applications, that cycle between higher inspiratory and lower expiratory pressures.[11,25] These ventilators have been used widely in acute settings because of their ease of administration and low cost, but they have been limited by a lack of alarms, monitoring capabilities, and $O_2$ blenders. Newer bilevel devices have been designed specifically for acute applications of NIPPV. They have features aimed at enhancing leak compensation and patient comfort, such as adjustable triggering and cycling mechanisms and rise times (the time to reach the preset inspiratory pressure). In addition to oxygen blenders, they also include graphic interfaces comparable to standard critical care ventilators, battery backup for in-hospital transport, and a variety of modes including proportional assist ventilation.

"Critical care" ventilators are those designed for invasive ventilation and, by virtue of microprocessor technology, offer a wide variety of modes, extensive alarm and monitoring capabilities, and $O_2$ blenders. In the past, these devices have been limited by triggering of nuisance alarms and limited leak-compensating abilities when used for NIPPV.[338] Many, however, now offer "NIV modes" that use a PSV mode, silence alarms, and add leak compensation and algorithms that facilitate triggering and cycling, even in the face of leaks. A laboratory study comparing bilevel ventilators with a critical care ventilator found that triggering, cycling, and leak-compensatory mechanisms were superior in several of the bilevel ventilators.[339] A more recent laboratory evaluation of NIV modes on critical care ventilators observed that most require additional adjustments in the face of leaks to function adequately.[340]

Because they use a single tube for both inspiration and expiration, bilevel ventilators contribute to $CO_2$ rebreathing unless used with a nonrebreathing exhalation valve that may increase expiratory resistance and expiratory work of breathing.[341,342] This rebreathing can be minimized by using an expiratory pressure greater than 4 cm $H_2O$[341,342] and masks with an in-mask exhalation valve situated over the bridge of the nose.[331,332] Comparisons of bilevel and critical care ventilators in intubated patients have demonstrated that gas exchange is equivalent, but work of breathing is increased during bilevel ventilation if minimal expiratory pressure levels (2 to 3 cm $H_2O$) are used.[344] When expiratory pressures of 5 cm $H_2O$ are used, however, the two types perform equally well in supporting gas exchange and reducing work of breathing, presumably because of counterbalancing of auto-PEEP. For delivery of NIV, clinical outcome studies using bilevel ventilators report success rates that compare favorably with those for critical care ventilators,[60,62] although no randomized, controlled trials have compared the two directly. Thus, the selection of either system can be justified, and the choice is often based on availability and financial considerations. Further, recent developments as described above have blurred the distinctions between the various types of ventilators.

### Chronic

Blower- or turbine-based portable pressure-limited bilevel ventilators are used most often in the long-term setting to deliver NIPPV. In a 9-year Swiss survey,[230] volume-limited devices predominated initially, but pressure-limited devices accounted for more than 90% of ventilator applications during the latter years. This shift has been driven by the low cost, ease of use, and portability of the pressure-limited devices. In addition, manufacturers have been steadily adding features that enhance monitoring. Some now offer wireless Internet connections that permit home monitoring of respiratory rate, oximetry, airway pressures, tidal volumes, and air leaks. Portable volume-limited positive-pressure ventilators are still preferred by some clinicians for specific applications. Because they offer sophisticated monitoring, they are used in patients requiring nearly continuous ventilator support. Because of their high pressure-generating capabilities, they may be preferred in patients with high respiratory system impedances, such those with morbid obesity or chest wall deformity, or they may be used to "stack" breaths to attain a higher inspired lung volume to increase cough flows.[241] Also, because volume-limited ventilators usually are driven by intermittent piston action rather than continuously operating blowers, backup battery life can be considerably longer. On the other hand, pressure-limited ventilators compensate for leaks more effectively than do volume-limited ventilators, although the compensatory flow goes, not surprisingly, mainly into "feeding" the leaks.[345] Studies in the long-term setting have not shown consistently better outcomes with one type of ventilator over the other,[346,347] however, and the choice usually becomes one of clinician and/or patient preference. In addition, a number of more recently introduced ventilators are "hybrid" devices that have the capability of delivering both pressure-limited and volume-limited breaths.

## Selection of a Ventilator Mode

### Acute

Although no studies have demonstrated superior efficacy in terms of avoiding intubation or mortality of one ventilator mode over another in the acute setting, some practitioners have found enhanced patient comfort or compliance with PSV.[348,349] Thus, although either volume-limited or pressure-limited modes can be used with the expectation of similar rates of success, pressure-limited modes appear to be accepted more readily by patients and are more commonly used (>90% of applications in some studies).[315] Some newer hybrid ventilators designed specifically for NIV are able to deliver both volume-limited or pressure-limited modes, with the capability of adjusting triggering and cycling sensitivity, rise time, and inspiratory duration to optimize patient comfort.[350] Proportional-assist ventilation, a unique mode that tracks instantaneous patient airflow, is capable of closely matching patient breathing pattern and hence potentially enhancing synchrony and comfort (see Chapter 13).[351] The flow signal is fed back to the ventilator as a raw signal (flow assist) or integrated over time (volume assist). Gains are imposed on both these signals and on a composite signal (proportional assist) that can be adjusted to assist a "proportion" of the patient's breathing effort. Theoretically, flow and volume assist are adjusted to match resistive and elastic work, respectively, which must be measured. These specific adjustments, however, are unnecessary when proportional-assist ventilation is used to deliver NIPPV clinically. Proportional-assist ventilation has been shown to be as effective and a more comfortable means of administering NIPPV than PSV delivered via a critical care ventilator[352] or a bilevel device.[353]

### Chronic

In the long-term setting, pressure-support and volume-limited ventilators achieve similar levels of overnight oxygenation.[347] Thirty consecutive patients, mainly with restrictive forms of chronic respiratory failure, received nasal volume-limited ventilation for 1 month followed by pressure-limited ventilation.[346] Only two patients failed to improve with volume-limited ventilation, whereas ten had increased $Pa_{CO_2}$ or symptomatic deterioration when switched to pressure-limited ventilation. Conversely, ten patients in another study had improved daytime blood gases when switched from volume-limited to pressure-limited ventilation.[354] Although these were not prospective, randomized trials, they show no clear advantage of one ventilator mode over the other. Thus, the choice between the two hinges on clinician preference and consideration of specific ventilator properties such as portability, pressure-generating capabilities, backup-battery life, ability to stack breaths, and other factors. In general, though, volume-limited ventilators have greater pressure-generating and alarm capabilities. A more recent approach has been to use hybrid modes such as volume-assured pressure preset ventilation that automatically adjusts the inspiratory pressure within preset limits to achieve a target minute volume. Thus far, the mode appears to function as well as standard pressure preset modes and may lower $CO_2$ levels more in patients with obesity hypoventilation than with standard bilevel therapy,[355] but not in patients with COPD and hypercapnia.[356] Similarly, a randomized crossover trial of twenty patients with restrictive thoracic disease showed no advantage of an autotitrating bilevel mode over a standard bilevel mode.[357]

## Ventilator Settings

### Acute

Two strategies have been described: the *high-low approach,* which starts with a higher inspiratory pressure (20 to 25 cm $H_2O$) and lowers it if patients are intolerant,[17] and the *low-high approach,* which starts with a low inspiratory pressure (8 to 10 cm $H_2O$) and raises it gradually as tolerated by the patient.[60] The former approach prioritizes rapid alleviation of respiratory distress; the latter aims to optimize patient comfort in an effort to maximize patient tolerance. Paramount with both approaches is the realization that subsequent adjustments are necessary depending on patient response. Higher initial pressure often must be adjusted downward, and it is very important that low initial pressure be raised (usually to 12 to 20 cm $H_2O$) within the first hour, if possible, to provide adequate ventilator assistance. The high-low approach may be preferable in patients with hypoxemic respiratory failure who often have high minute volumes and are very dyspneic. L'Her et al[358] demonstrated that in patients with acute lung injury treated with NIPPV, higher pressure support levels were effective at relieving dyspnea while increases in PEEP were effective at improving oxygenation.

Expiratory pressure (or PEEP) is used routinely with bilevel ventilators and is optional with volume-limited ventilators. Bilevel ventilators require a bias flow during expiration to flush $CO_2$ from the single ventilator tube and avoid rebreathing.[341] Minimal expiratory pressure with these ventilators is in the 3 to 4 cm $H_2O$ range. Higher expiratory pressures (typically 4 to 8 cm $H_2O$) are used to counterbalance intrinsic PEEP during treatment of exacerbations of COPD or to enhance oxygenation. It is important to recall that the difference between inspiratory and expiratory pressure is the level of PSV, so inspiratory pressure must be increased in tandem with expiratory pressure if the same level of ventilatory assistance is to be maintained. Adjusting the rate of pressurization (or rise time) may be useful to enhance comfort; patients with COPD prefer slightly more rapid rise times than restrictive patients. Very rapid pressurization rates minimize the work of breathing in patients with COPD but may be sensed as less comfortable by patients than slightly lower pressurization rates.[359]

### Chronic

No consensus has been reached on how to select ventilator settings for patients in the long-term setting. In the sleep laboratory, one approach is to increase expiratory pressure until apneas are eliminated and inspiratory pressure until hypopneas are eliminated without inducing excessive arousals. The American Academy of Sleep Medicine has proposed guidelines for achieving this titration.[360] This approach, however, is based largely on expert opinion and does not ensure that the pressures selected will alleviate hypoventilation, nor does it facilitate initial adaptation if the patient finds the recommended initial pressures intolerable. Thus, I prefers a gradual up titration of pressures over weeks or even months as tolerated by the patient. Initial inspiratory pressure is 6 to 10 cm $H_2O$, with increases weekly by 1 to 2 cm $H_2O$ as tolerated. Expiratory pressure is set at 3 to 4 cm $H_2O$ and rarely is increased above 6 cm $H_2O$ unless sleep apnea is deemed an important contributor. For volume-limited ventilation, an initial tidal volume of 10 to 15 mL/kg has been recommended, in excess of the standard recommendations for invasive ventilation, because of the need to compensate for air leaks.[361] Parreira et al[362] found that a tidal volume of 13 mL/kg optimized assisted minute volume in a group of patients with restrictive thoracic disorders.

A backup rate sufficiently high to control breathing nocturnally has been recommended for patients with neuromuscular disease to maximize respiratory muscle rest and prevent apneas. In patients with severe stable COPD, on the other hand, the need for a backup rate is not clear, considering that one controlled trial of NIPPV found significant benefit using a spontaneous ventilator mode without a backup rate.[332] Compared with a spontaneous mode, these authors found that use of a backup rate had no effect on nocturnal gas exchange in patients with COPD and chronic respiratory failure. On the other hand, Parreira et al[362] found that minute volume was optimized when patients with restrictive thoracic disorders used a relatively high backup rate of 23 breaths/min.

## Adjuncts to Noninvasive Ventilation

With bilevel ventilators, supplemental $O_2$ can be provided directly through tubing connected to a nipple in the mask or to a T connector in the ventilator tubing, with liter flow adjusted to keep $SaO_2$ above 90% to 92%. Maximal $FI_{O_2}$ using this setup is only 45% to 50%. $FI_{O_2}$ delivered via bilevel ventilators depends on a number of factors, including $O_2$ flow rate, breathing pattern, ventilator settings, and location of the $O_2$ connection (connection to the mask gives a higher $FI_{O_2}$).[363] With critical care ventilators and some bilevel devices designed for acute applications, standard $O_2$ blenders are used to accurately provide the desired. Thus, these latter ventilators are preferred for patients with hypoxemic respiratory failure.

Humidification may enhance comfort during NIPPV and is advised if NIPPV is to be used for more than a few hours, although effects on NIV failure rates have not been established.[364] A heated humidifier is preferred over a heat and moisture exchanger because the latter adds to work of breathing[365] and may interfere with triggering and cycling. Also, with excessive air leaking, a heated humidifier lowers nasal resistance.[23] In the long-term setting, humidification usually is provided, particularly during the winter months in colder climates. Nasogastric tubes are not recommended routinely as adjuncts to NIV, even when oronasal masks are used.

## Noninvasive Techniques to Assist Cough

Because it provides no direct access to the lower airways, as does invasive ventilation, NIPPV depends on the integrity of airway protective reflexes for its success. In the acute setting, patients with excessive secretions or severe cough impairment are intubated rather than managed noninvasively. In the long-term setting, however, a number of techniques have been developed to enhance secretion removal in patients with compromised secretion-clearance capabilities. These techniques are of greatest value in patients with neuromuscular disease and weakened expiratory muscles. Severe bulbar involvement, such as occurs with ALS, is treated most effectively with invasive ventilation. Secretion retention related to abnormal mucus, such as occurs with cystic fibrosis, is beyond the scope of this chapter.

An effective cough depends on the ability to generate adequate expiratory airflow, estimated at more than 160 L/min.[366] Expiratory airflow is determined by lung and chest wall elasticity, airway conductance, and at least at higher lung volumes, expiratory muscle force. By generating an adequate vital capacity (> 2.5 L) to take advantage of respiratory system elasticity, inspiratory muscle function also contributes to cough adequacy. In addition, an effective cough requires the ability to close the glottis so that explosive release of intrathoracic pressure can generate high peak expiratory cough flows.[367] When patients with severe neuromuscular disease are too weak to take advantage of these mechanisms and have insufficient cough flows, techniques to assist cough should be applied.

The simplest maneuver to augment cough flow is manually assisted or “quad” coughing. This consists of firm, quick thrusts applied to the abdomen using the palms of the hands, timed to coincide with the patient’s cough effort.[241] The technique should be taught to caregivers of patients with severe respiratory muscle weakness with instructions to use it whenever the patient has difficulty expectorating secretions. With practice, the technique can be applied effectively and frequently with minimal discomfort to the patient. Peak expiratory flows can be increased severalfold when manually assisted coughing is applied successfully.[368] To minimize the risk of regurgitation and aspiration of gastric contents, the patient should be semi-upright when manually assisted coughing is applied, and the technique should be used cautiously after meals. The technique can be used, though, in patients with gastric feeding tubes.

Although manually assisted coughing may enhance expiratory force, it does not augment inspired volume. Thus, patients with severely restricted volumes may not achieve sufficient cough flows even when assisted by skilled caregivers. To overcome this problem, the inhaled volume should be augmented. One approach is to “stack” breaths using glossopharyngeal breathing[368] or volume-limited ventilation and then to augment the cough using manual assistance. Another is to use a mechanical insufflator–exsufflator, a device that was developed during the polio epidemics to aid in airway secretion removal.[368] This device delivers a positive inspiratory pressure of 30 to 40 cm $H_2O$ via a face mask and then rapidly switches to an equal negative pressure. The positive-pressure ensures delivery of an adequate tidal volume, and the negative pressure has the effect of simulating the rapid expiratory flows generated by a cough. Use of the insufflator–exsufflator has been combined with manually assisted coughing in an effort to further augment cough flows.[369]

The mechanical insufflator–exsufflator increases cough flows in patients with neuromuscular weakness but is less effective in patients with kyphoscoliosis, and in one study, actually decreased cough flows of patients with COPD.[370] In another study, however, mechanical insufflation–exsufflation decreased dyspnea and improved oxygenation not only in neuromuscular patients but also in patients with airflow obstruction.[371] Although no controlled trials have evaluated the efficacy of the cough insufflator–exsufflator, anecdotal evidence suggests that it enhances removal of secretions in patients with impaired cough. It has been reported to reduce failures (need for intubation) in neuromuscular patients in critical care settings,[372] reduce the occurrence of atelectasis and pneumonias in children with neuromuscular disease,[373] and improve cough flows in patients with ALS unless there is bulbar dysfunction, which may predispose to upper airway collapse.[374] It has been particularly useful in patient homes to treat episodes of acute bronchitis, permitting avoidance of hospitalization.[375] One recent European study demonstrated cost savings by staying in telephone contact with patients and using the cough insufflator–exsufflator on an “on-demand” basis to treat exacerbations.[376] Other devices that aid expectoration, such as the percussive ventilator, Hayek oscillator, and vibratory vest, have some theoretical advantages over other techniques for assisting secretion removal.[241] Their use of high-frequency vibrations (up to 15 Hz) may facilitate mobilization of airway secretions. Unfortunately, even anecdotal evidence to support their use is lacking.

Clinicians caring for patients with severely impaired cough should be familiar with the various techniques available to assist expectoration. These are particularly important with NIV because there is no direct access to the airway, and secretion retention is a frequent complication and common cause for failure. Although controlled data are lacking, these techniques appear to help in maintaining airway patency in patients with cough impairment during use of NIV in both acute and chronic settings.

## Role of the Clinician: Time Demands, Importance of Experience, and Guidelines

An experienced clinician conveying an air of assuredness to patients is thought to be crucial to the success of NIPPV. The clinician should motivate the patient, explaining the purpose of each piece of equipment and preparing the patient for each step in the initiation process. Patients should be reassured, encouraged to communicate any discomfort or fears, and coached in ways to coordinate their breathing with the ventilator. When using nasal masks, patients are instructed to keep their mouths shut.

Time demands on medical personnel have been a concern for the delivery of NIPPV. Chevrolet et al[377] were the first to draw attention to this potential problem, reporting large time demands on nurses during administration of NIPPV. Conversely, Kramer et al[60] found that compared with controls, NIPPV patients tended to require more time from respiratory therapists during the first 8 hours, an amount that fell significantly during the second 8 hours. Nava et al[378] also found that respiratory therapists spent more time during the first 48 hours caring for NIPPV patients than invasively ventilated patients. These findings indicate that NIPPV initially requires more time to administer than conventional therapy, for interface fitting and initial ventilator adjustment, although these demands diminish rapidly after the first few hours. Nurses, respiratory therapists, physicians, or some combination of these must spend the additional time, depending on institutional practices.

As might be anticipated, the experience of personnel also appears to be important for the success of NIPPV. Girou et al[15] reviewed the experience of a twenty-six-bed French ICU between January 1994 and December 2001 on 479 patients with either COPD or cardiogenic pulmonary edema requiring ventilator assistance, invasively or noninvasively. Use of NIV increased from approximately 20% to 90% (of the patients) during the course of the study, associated with a decrease in the rate of nosocomial pneumonias from 20% to 8% and in ICU mortality from 21% to 7% (all $p < 0.05$). The authors speculated that a “learning effect” over the course of the study was responsible for the improved outcomes. Over an 8-year period in their ICU, Carlucci et al[379] found that NIPPV success rates increased in patients with COPD despite a worsening of the severity of illness as staff gained experience with the technique.

Whether or not guidelines for NIPPV implementation can improve patient outcomes remains to be established. Sinuff et al[380] found that clinician behavior changed after implementation of a guideline at their single academic institution, but overall patient outcomes were not altered. More recently, Kikuchi et al[381] observed a decreased mortality after institution of a NIPPV protocol for acute respiratory failure, although the before/after study design cannot control for other changes related to passage of time.

# Monitoring

Patients receiving NIV are monitored to determine whether the goals are being achieved (see Table 18-8).

## Subjective Responses

The key aims of NIPPV are alleviation of respiratory distress in the acute setting and of fatigue, hypersomnolence, and other symptoms of impaired sleep in the chronic setting while achieving patient tolerance. Agitation and mask discomfort are challenges during NIPPV. These aspects can be assessed easily using bedside observation and patient queries. Some patients minimize or deny discomfort and still may have great difficulty adapting successfully to NIV, so clinicians not should only query patients but also should observe for nonverbal signs of distress or discomfort.

## Physiologic Responses

Evidence of physiologic improvement within the first 2 hours, including decreases in respiratory and heart rates, diminished sternocleidomastoid muscle activity, and elimination of abdominal paradox, portends a favorable NIPPV outcome.[133,196] Patients should be breathing in synchrony with the ventilator, and air leaking should be minimal. Some clinicians also monitor tidal volumes, aiming for delivered volumes in excess of 7 mL/kg.[382] Relying on ventilator monitoring to follow tidal volumes may be misleading, however, particularly during use of bilevel ventilators, because these integrate the inspired flow signal and may be very inaccurate in the face of air leaks.

## Gas Exchange

Improvement in gas exchange as determined by continuous oximetry and occasional blood gases is a key aim in acute application of NIPPV, although improvement in ventilation may occur gradually over hours.[17] In chronic stable patients, the improvement in daytime gas exchange occurs even more slowly, over a period of weeks, depending on the duration of daily ventilator use. Some patients adapt slowly and require up to several months before they sleep through the night using the ventilator. Arterial blood-gas measurement should be delayed until the patient is consistently using the ventilator for a period of time likely to improve gas exchange: usually at least 4 to 6 hours a day. No consensus on an ideal level for daytime $Pa_{CO_2}$ has been reached; most investigators target levels in the middle 40s. In my experience, a daytime $Pa_{CO_2}$ of up to 60 mm Hg, or even higher, may be tolerated without hypersomnolence or evidence of cor pulmonale as long as oxygenation is adequate. Noninvasive $CO_2$ monitoring may be useful for trending purposes in patients with normal lung parenchyma such as those with neuromuscular disease. Transcutaneous $Pa_{CO_2}$ is probably the most useful because variable air leaks and breathing patterns and dilution secondary to bias flow with some ventilators render end-tidal $CO_2$ recordings inaccurate, particularly if the patient has parenchymal lung disease.[383]

## Sleep

Little is known about sleep during NIPPV in the acute setting, and the role of sleep monitoring in the long-term setting is controversial. As noted earlier, some clinicians prefer to use the sleep laboratory to decide on initial settings for NIPPV. Others begin most patients, particularly those with neuromuscular disease, without a polysomnographic evaluation. If both approaches prove to be equal in achieving the goals of NIV, it would be difficult to argue that routine sleep studies are necessary. The relative utilities of polysomnography, multichannel portable recordings, and nocturnal oximetry are also unclear in follow-up of long-term NIPPV, but one pragmatic approach is to screen patients using home oximetry and to perform more sophisticated studies when the oximetry results indicate the need for further evaluation.

## Adaptation

### Acute

It is critically important to ascertain that the delivered pressures are sufficient to alleviate respiratory distress; a common mistake is to fail to increase pressures quickly enough, and the patient fails because of inadequate ventilator support. The patient also should use the ventilator for more time initially, with increasing periods of time off the ventilator as the underlying condition improves. Some clinicians encourage use most of the time initially, as dictated by the degree of respiratory distress during ventilator-free intervals.[60] Others employ sequential use,[384] wherein periods of use alternate with lengthy ventilator-free periods; total daily use averages only 6 hours, although this would be suitable only for mildly ill patients. When no respiratory distress recurs during ventilator-free intervals, ventilator assistance is discontinued. Total duration of ventilator assistance depends on the speed of resolution of the respiratory failure. Patients with acute pulmonary edema require an average of 6 to 7 hours of ventilator use,[114] whereas patients with COPD average 2 days or more.[60] Some patients may continue nocturnal ventilation after discharge from the hospital, following guidelines for long-term use of NIPPV.

### Chronic

Patients start more gradually and increase periods of use as tolerated. Anxious patients may begin with only 1 to 2 hours of daytime use followed by gradually increasing periods of nocturnal use over several weeks or even months. Compared with the acute setting, urgency in the chronic setting is less, and because the intent is to use the ventilator during sleep, great care must be exercised in optimizing patient comfort. During this period, visits from a home respiratory therapist are helpful to assess comfort and address any problems that arise. Criner et al[278] found that 36% of patients required further adjustments in mask or ventilator settings, even after discharge from a several-week stay in an inpatient ventilator unit.

## Adverse Effects and Complications

In properly selected patients, NIPPV is safe and well tolerated, and most of the adverse effects are related to the interface or ventilator (Table 18-9). Approximately 10% to 15% of patients fail to tolerate interfaces despite adjustments in strap tension, repositioning, and trials of different sizes and types of interfaces. These patients may have claustrophobia or high levels of anxiety and fail even after multiple attempts at mask readjustment and judicious use of sedation. Other mask-related adverse effects include erythema, pain, and ulceration on the bridge of the nose. Minimizing strap tension and advances in mask technology, with softer silicon seals and routine use of artificial skin on the bridge of the nose, are associated with less frequent nasal bridge ulceration, which had been as high as 40% in earlier studies.[385]

**Table 18-9: Frequency of Adverse Side Effects and Complications of Noninvasive Positive-Pressure Ventilation with Possible Remedies**

| Complication | Occurrence (%)* | Possible Remedy |
|---|---|---|
| **Mask-related** | | |
| Discomfort | 30% to 50% | Check fit, adjust strap, new mask type |
| Facial skin erythema | 20% to 34% | Loosen straps, apply artificial skin |
| Claustrophobia | 5% to 10% | Smaller mask, sedation |
| Nasal bridge ulceration | 5% to 10% | Loosen straps, artificial skin, change mask type |
| Acneiform rash | 5% to 10% | Topical steroids or antibiotics |
| **Air pressure or flow related** | | |
| Nasal congestion | 20% to 50% | Nasal steroids, decongestant/antihistamines |
| Sinus/ear pain | 10% to 30% | Reduce pressure if intolerable |
| Nasal/oral dryness | 10% to 20% | Nasal saline/emollients, add humidifier, decrease leak |
| Eye irritation | 10% to 20% | Check mask fit, readjust straps |
| Gastric insufflation | 5% to 10% | Reassure, simethicone, reduce pressure if intolerable |
| **Air leaks** | 80% to 100% | Encourage mouth closure, try chin straps, oronasal mask |
| | | If using nasal mask, reduce pressures slightly |
| **Major complications** | | |
| Aspiration pneumonia | < 5% | Careful patient selection |
| Hypotension | < 5% | Reduce pressure |
| Pneumothorax | < 5% | Stop ventilation if possible, reduce pressure; if not, use thoracostomy tube if indicated |

*Occurrences estimated from literature and my experience.

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Air pressure-related and flow-related adverse effects include oronasal dryness or congestion that may respond to humidification or decongestants, sinus and ear pain, eye irritation from air leakage under the mask seal on the sides of the nose, and gastric insufflation. Readjusting the mask seal to reduce air leaking or reducing inspiratory pressure, if possible, may help.

Air leaking is ubiquitous during NIV, either under the seal or through the mouth with nasal ventilation. Air leaking adds to discomfort and may interfere with ventilator triggering and cycling, as well as efficacy of ventilation. The leaks usually are tolerated as long as the ventilator compensates adequately, as most bilevel ventilators do. As discussed earlier, bilevel ventilators cannot function properly without a small intentional leak in the tubing, which is necessary for removal of $CO_2$ to prevent rebreathing. Most volume-limited modes compensate poorly for leaks, but large air leaks may compromise the effectiveness of any form of NIPPV. Attempts to control air leaks should start with a reassessment of mask fit and readjustment of the head straps. To reduce air leaking through the mouth during nasal ventilation, edentulous patients should not be treated with nasal masks, others are coached to keep their mouths shut, chin straps may be used, or an oronasal mask may be tried.

Major complications, such as pneumothoraces, are unusual, probably because inflation pressures are low compared with those used with invasive ventilation. Lack of patient cooperation interferes with efficacy and may be ameliorated by judicious use of sedation, such as low doses of benzodiazepines. Unremitting agitation should be considered an indication for intubation. Aspiration is a reported complication,[386] but should be unusual if patients with swallowing dysfunction and problems clearing secretions are excluded. Routine insertion of nasogastric tubes has been recommended at some centers to lower the risk of aspiration during use of face masks,[382] but there are no data available to support this practice, and it is no longer recommended. Progressive hypoventilation occurs in a small minority of patients, usually necessitating intubation. Uncooperativeness, lack of synchronization with the ventilator, inability to tolerate adequate inflation pressures, and excessive air leaking are common causes for this predicament, and measures aimed at correcting these may be helpful. Rarely, nasal obstruction contributes and may respond to decongestant sprays.

In the acute setting, NIPPV fails in roughly a third to a quarter of patients, depending on many factors, including skill and experience of the team, occurrence of adverse effects and complications as discussed earlier, and underlying severity of the patient's illness. Progression of the underlying process, such as worsening hypoxemia, also may be responsible for failure. Close monitoring with proactive efforts to address and minimize adverse effects should minimize failure rates.

## Summary and Conclusions

In the acute setting, evidence now supports NIPPV in the treatment of respiratory failure secondary to acute exacerbations of COPD, acute cardiogenic pulmonary edema (which can be managed with CPAP as well), and immunocompromised states and to facilitate weaning in patients with COPD. Weaker evidence supports NIPPV in the treatment of other forms of respiratory failure, including respiratory insufficiency in patients with asthma, post–lung resection, or those who decline intubation. NIPPV should not be used routinely in patients with ARDS or severe pneumonia. Regardless of the underlying cause of the respiratory failure, patients to receive NIPPV must be selected carefully. Those with mild deteriorations likely will succeed without ventilator assistance, and prompt intubation usually is preferred in severely ill patients. Patients with unstable medical conditions, inability to protect the airway or clear secretions, or uncooperativeness are excluded from consideration. Patients selected according to these guidelines and monitored closely can be managed with NIPPV with the expectation that intubation and its inherent complications will be avoided.

In the chronic setting, NIPPV has long been considered the modality of first choice for patients with respiratory failure secondary to neuromuscular disease or chest wall deformity. Randomized, controlled trials have not been done because of ethical concerns, but the ability of NIPPV to improve symptoms, gas exchange, quality of life, and survival in these patients is widely accepted. NIPPV for patients with severe stable COPD has been more controversial because of conflicting data, but it may prevent deterioration of gas exchange, improve quality of life, and reduce the need for hospitalization in patients with severe $CO_2$ retention and nocturnal hypoventilation who adhere to therapy (which is often not the case). In some countries, the obesity-hypoventilation syndrome has become the largest diagnostic category for NIPPV at home, perhaps because of increasing recognition and the obesity epidemic.

In both the acute and chronic settings, experience and skill with the implementation of NIPPV are important. Proper fit and application of the mask are keys to success. Although the type of ventilator is not as important, ventilator mode and settings affect comfort and effectiveness. In carefully selected patients initiated according to current recommendations and monitored in an appropriate setting, NIPPV can be delivered safely with no or minor adverse effects. The failure rate remains substantial at roughly a quarter to a third of patients, but hopefully will decline as clinicians gain experience, selection criteria are refined, and technology improves. NIPPV has assumed an important role in acute and chronic care settings and should be considered a lung-protective strategy for certain forms of respiratory failure. In addition, clinicians caring for patients with respiratory failure should have the necessary skill and experience with NIPPV applications.

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